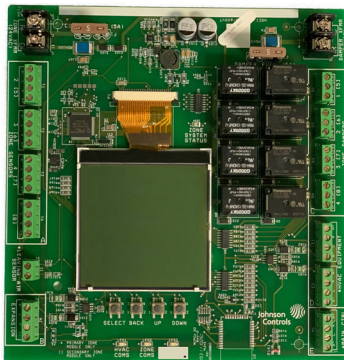


INSTALLATION MANUAL

Hx 3 ZONE CONTROL MODULE

MODELS: S1-ZMC401A



LIST OF SECTIONS

General information	2	Main Control and zone thermostats	39
Hx 3 Zone Control Module kit	2	Hx 3 Zone Control Module	39
Safety considerations	2	Hx 3 Zone Control Module - LCD and features	41
Understanding the Hx 3 Zone Control Module	3	Using the LCD	41
Limitations	3	Incomplete profile	41
Hx 3 Touch Screen Thermostat software requirements	3	Unassigned zone sensor	42
Hx 3 Zone Control Module layout	4	Component miswire	42
LED and LCD functionality	9	General display	43
Location	14	Active faults	43
Mounting the Hx 3 Zone Control Module	14	Software updates	44
Wiring	15	Zone settings	44
Zoning system installation	19	Sensor status - Primary Zone Module only	44
Part 1: Prepare for zoning system installation	19	System status - Primary Zone Module only	45
Part 2: Install the zoning system components	20	Auto detection - Primary Zone Module only	45
Part 3: Complete initial power-up	21	Configurations - Primary Zone Module only	46
Part 4: Configure equipment and zoning components	21	Test mode - Primary Zone Module only	46
Part 5: Configure settings	25	Zone system faults	50
Part 6: Instruct the homeowner	28	Restore defaults	51
Zoning system settings	28	Repair	52
Accessing zone settings	28	Restoring system defaults	52
Heat/cool changeover time	28	Configuring zone thermostat addresses	52
Damper timing	29		
Airflow bleeding	29		
Maximum airflow for individual zones	32		
Zone names	37		
Zone safeties	38		
Zoning system software updates	38		

Section I: General information

The Hx™3 Zone Control Module kit is for use in a multi-zone HVAC system. Multi-zone systems can save energy and provide optimum comfort in a site. The kit includes the Hx 3 Zone Control Module and the leaving air sensor. The main function of the Hx 3 Zone Control Module is to communicate with the Hx™3 Touch Screen Thermostat acting as the Main Control in the zoning system, and provide 24 VAC outputs to the zone dampers.

The purpose of this document is to do the following:

- Explain how to install and use the Hx 3 Zone Control Module, and how it operates within the zoning system.
- Provide information on the wider zoning system, including a procedure for zoning system installation.

For more information on specific zoning components and indoor and outdoor equipment, refer to the associated documentation.

The zoning system can include different zoning components, depending on the requirements of a site and the number of zones. Components have specific names in the zoning software. Table 1 provides an overview. The product naming used throughout this document reflects Table 1.

Table 1: Zoning components

Component	Function	Name in zoning software	Notes
Hx 3 Touch Screen Thermostat (S1-THXU430W)	Act as Main Control (zone 1)	Main Control	You must use a Hx 3 Touch Screen Thermostat as the Main Control for zoning operation. You can use additional Hx 3 Touch Screen Thermostats as zone thermostats in zones 2–8 as required.
	Act as zone thermostat sensor in zones 2–8	Zone Thermostat-Hx	
Hx™3 Zone Thermostat (S1-ZCC01A)	Act as zone thermostat sensor in zones 2–8	Zone Thermostat	You can use Hx 3 Touch Screen Thermostats, Hx 3 Zone Thermostats, or Hx 3 Zone Sensors as zone sensors in zones 2–8.
Hx™3 Zone Sensor (S1-TCZSN02)	Act as zone sensor in zones 2–8	Zone Sensor	You can use Hx 3 Touch Screen Thermostats, Hx 3 Zone Thermostats, or Hx 3 Zone Sensors as zone sensors in zones 2–8. The Hx 3 Zone Sensor senses zone temperature only.
Hx 3 Zone Control Module (S1-ZMC401A)	Act as Primary Zone Module (zones 1–4)	Primary Zone Module or Primary ZM	You must use a Hx 3 Zone Control Module (Primary Zone Module) for zoning operation. Zoning systems with more than four zones require a second Hx 3 Zone Control Module (Secondary Zone Module) for zones 5–8.
	Act as Secondary Zone Module (zones 5–8)	Secondary Zone Module or Secondary ZM	

IMPORTANT: Sound residential design practices require compliance with the following ACCA manuals: Manual J (Load Calculations), Manual S (Equipment Selection), Manual D (Duct Design), and Manual T (Air Distribution).

Hx 3 Zone Control Module kit

The following table shows the parts included in the Hx 3 Zone Control Module kit. Examine the kit to ensure all parts are included.

Table 2: Kit contents

Quantity	Description
1	Hx 3 Zone Control Module
4	Hx 3 Zone Control Module mounting screws
1	Installation Manual for installers and User's Information Manual for homeowners
1	Quick Reference Guide for the installation instructions
1	Limited Warranty
1	Leaving air sensor

Section II: Safety considerations

 This is a safety alert symbol. When you see this symbol on labels or in manuals, be alert to the potential for personal injury and equipment damage.

Understand and pay particular attention to the signal words **DANGER**, **WARNING**, and **CAUTION**.

DANGER indicates an **imminently** hazardous situation, which, if not avoided, will result in death or serious injury.

WARNING indicates a **potentially** hazardous situation, which, if not avoided, could result in death or serious injury.

CAUTION indicates a potentially hazardous situation, which, if not avoided may result in minor or moderate injury. It is also used to alert against unsafe practices and hazards involving only property damage.

Section III: Understanding the Hx 3 Zone Control Module

Limitations

NOTICE

Compatible Communicating Equipment Models
The Hx 3 Zone Control Module (S1-ZMC401A) is only compatible with the communicating equipment models listed below.
Furnaces: TM8V*, TMLV*, TM9V*, YPLC*, CPLC*, LPLC*, TPLC*, YP9C*, CP9C*, LP9C*, TP9C*
Air Handlers: AVC*, AVV*, MVC*
Air Conditioners: YXT*, AC19*, AL19*, YXV*, AC21*, AL21*
Heat Pumps: YZT*, HC19B*, HL19B*, YZV*, HC20*, HL20*

NOTICE

Compatible Communicating Zone Sensors
The Hx 3 Zone Control Module (S1-ZMC401A) is only compatible with the communicating zone sensors listed below.
Hx 3 Zone Control Module: S1-ZMC401A
Hx 3 Touch Screen Thermostat (Main Control and Zone Thermostat-Hx): S1-THXU430W
Hx 3 Zone Thermostat: S1-ZCC01A
Hx 3 Zone Sensor: S1-TCZSN02

The Hx 3 Zone Control Module kit requires at least one Hx 3 Touch Screen Thermostat (S1-THXU430W) to act as the Main Control. This product does not support direct connection with pneumatic dampers. Use 24 VAC three-wire power open/power close dampers.

NOTICE

The indoor unit must be capable of communications. Use of the communicating interface control (S1-33102953000) is not supported with zoning.

Hx 3 Touch Screen Thermostat software requirements

You must use a Hx 3 Touch Screen Thermostat as the Main Control. For easy installation, use Hx 3 Touch Screen Thermostats as the Main Control and zone thermostats (for zones 2–8) on the zoning system. It is best practice to update the thermostat software from non-zoning software to zoning software (02.00 or greater) before installation. Verify the software version shown on the outside of the thermostat box as shown in the following figure:



Figure 1: Hx 3 Touch Screen Thermostat box label

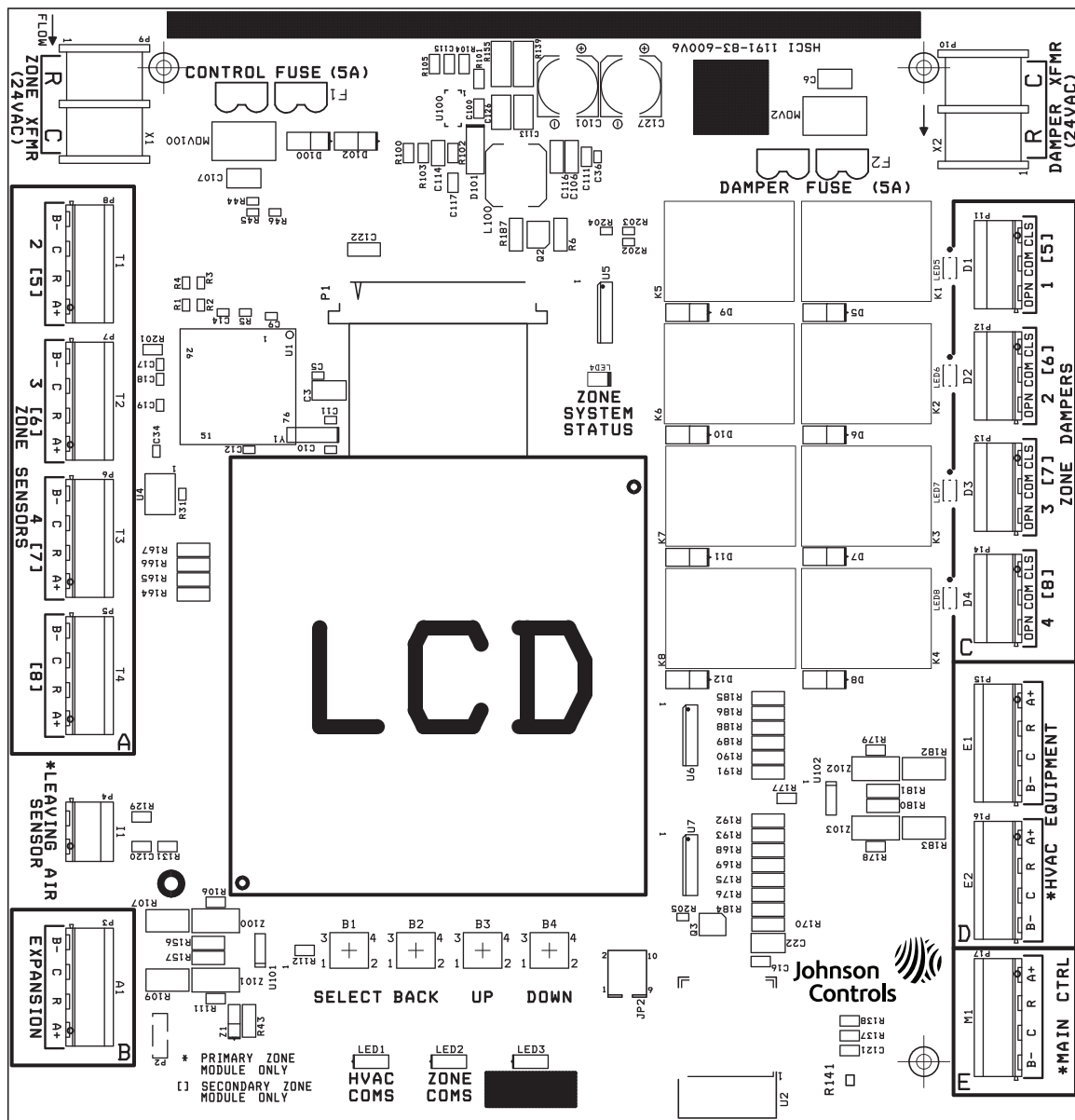
- IMPORTANT:** Confirm the software version on the outside label of the Hx 3 Touch Screen Thermostat box:
- When the thermostat is installed as the Main Control, the software version must be greater than 01.11 to support zoning operation.
 - When the thermostat is installed as a zone thermostat, the software version must be greater than 02.00 to support zoning operation.

Hx 3 Zone Control Module layout

Figure 2 shows the Hx 3 Zone Control Module from a front view. The following table gives an overview of the terminals on the Hx 3 Zone Control module, their location, and their use.

Table 3: Hx 3 Zone Control Module - terminals

Terminals	Location	Use	Notes
Four four-wire communication terminals	In Quadrant A on the left side	For connecting zone sensors	
One four-wire communication terminal	In Quadrant B in the lower left corner	For connecting an additional Hx 3 Zone Control Module (Secondary Zone Module) for zones 5–8	
Four three-wire damper terminals	In Quadrant C on the right side	For connecting 24 VAC power open/power close dampers	The Hx 3 Zone Control Module only supports power open/power close 24 VAC (three-wire) dampers. The zone damper output terminals are used to control damper positions commanded by the Hx 3 Zone Control Module.
Two four-wire communication terminals	In Quadrant D on the lower right side	For connecting the indoor and outdoor equipment controls	Attach the indoor and outdoor controls to the Primary Zone Module only.
One four-wire communication terminal	In Quadrant E in the lower right corner	For connecting the Hx 3 Touch Screen Thermostat acting as the Main Control	Attach the Main Control to the Primary Zone Module only.
One two-wire terminal	Between Quadrant A and Quadrant B on the left side	For connecting the leaving air sensor	Attach the leaving air sensor to the Primary Zone Module only.
One two-screw terminal	In the upper right corner	For the damper power transformer	Attach a separate damper power transformer. The damper power must not be the same as control power.
One two-screw terminal	In the upper left corner	For zone system power	The original equipment manufacturer (OEM) recommended power source for the zone system is the indoor equipment control transformer. The Hx 3 Zone Control Module input power must only be used if the indoor equipment transformer is isolated from the zone system power. If the indoor equipment R wire is connected in Quadrant D, do not connect an additional transformer to this connection.



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Figure 2: Hx 3 Zone Control Module board layout

Table 4: Zone damper terminal designations

Signal	Definition	Label
Damper close output	Damper power closed	CLS
Damper common output	Damper common	COM
Damper open output	Damper power open	OPN

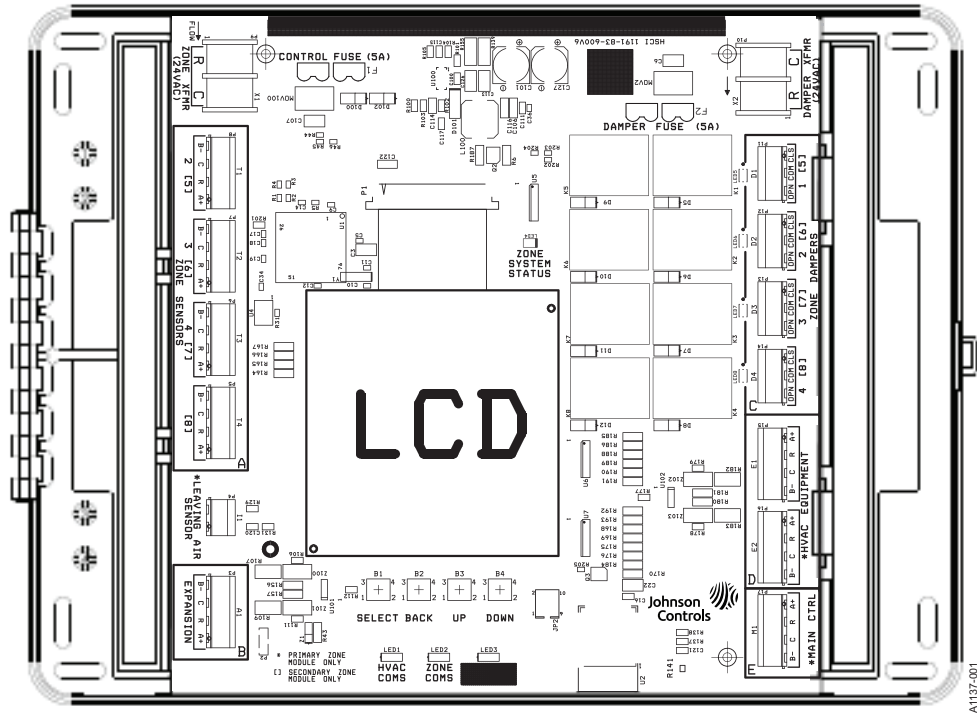
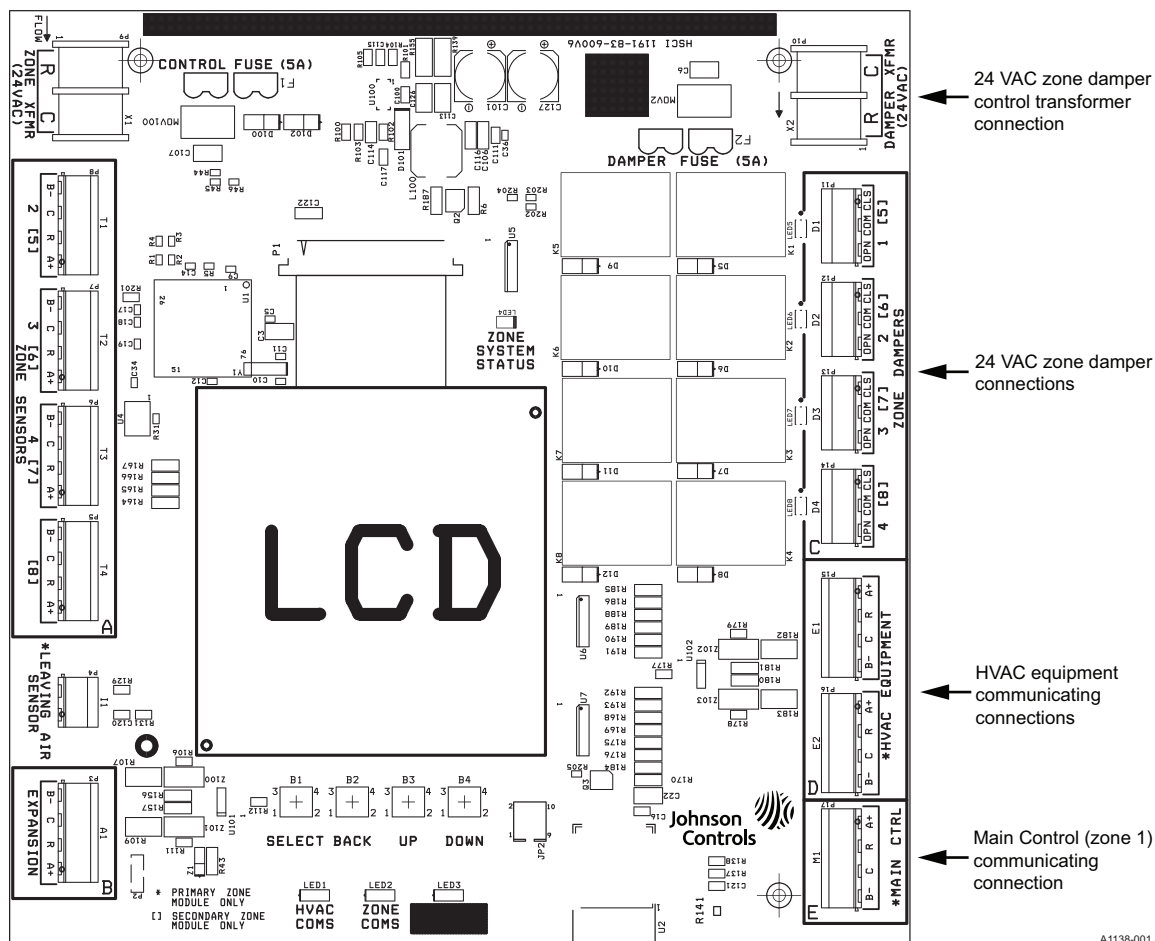


Figure 3: Hx 3 Zone Control Module



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Figure 4: Hx 3 Touch Screen Thermostat (Main Control), HVAC equipment input, damper power input, and zone damper outputs

Table 5: Communication terminal designations

Signal	Definition	Label
Data	Non-inverted signal	A+
Low-voltage power hot	24 VAC (hot)	R
Low-voltage power common and data ground	24 VAC (common)	C
Data	Inverted signal	B-

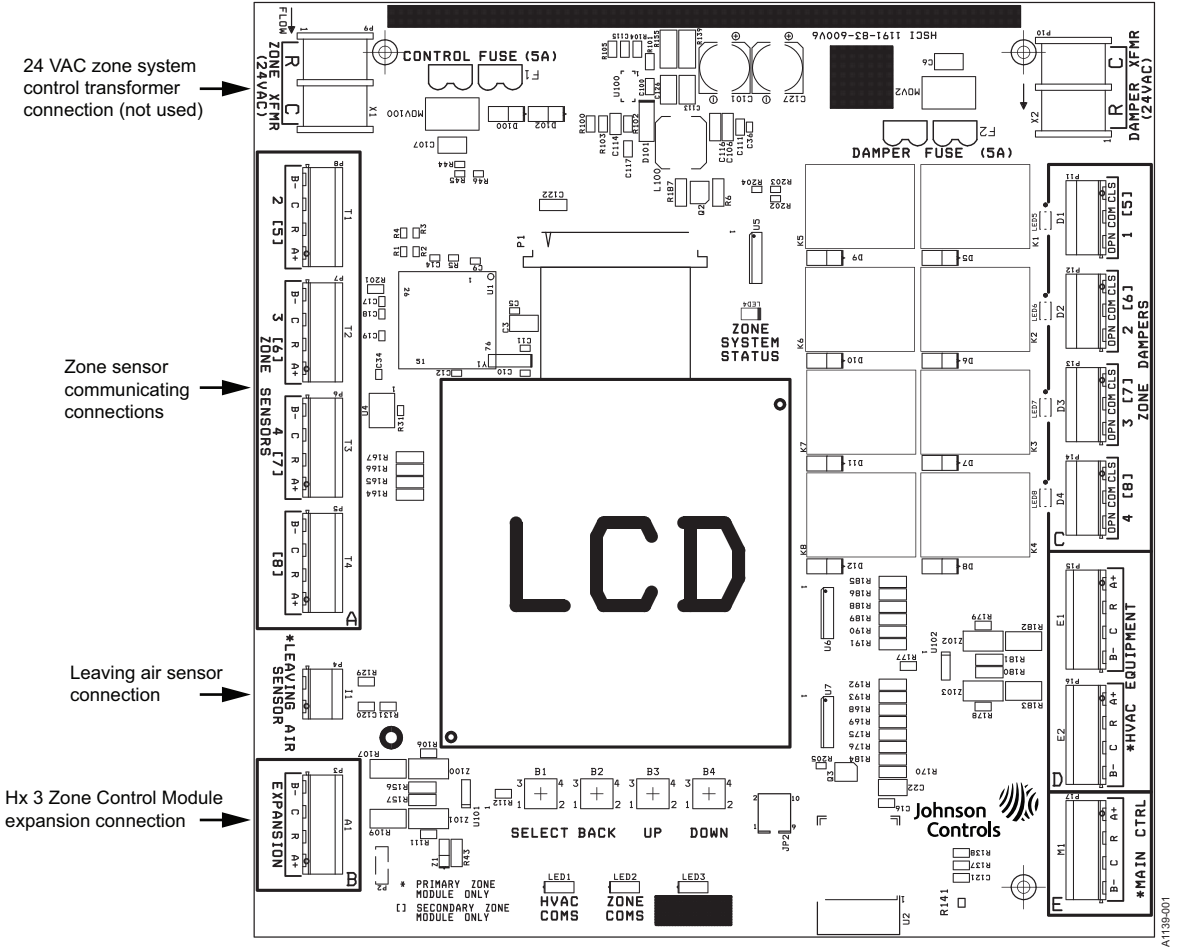


Figure 5: Zone sensors, leaving air sensor, and zone module expansion outputs

See Table 6 and Figure 6 for OEM recommended wire color, gauge, and terminations.

Table 6: Communication wire color

Wire color	Wire gauge	Label
Green	18 – Solid Copper	A+
Red	18 – Solid Copper	R
Black	18 – Solid Copper	C
White	18 – Solid Copper	B-

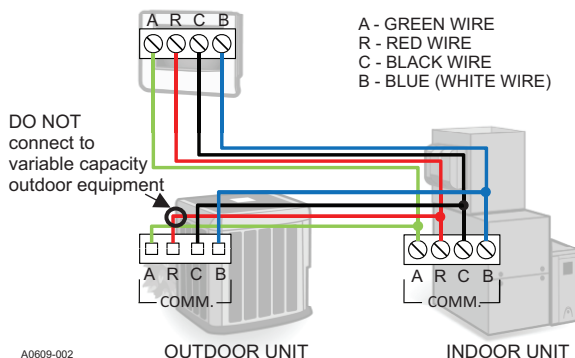


Figure 6: Communication wire terminals

LED and LCD functionality

The LCD and LEDs located on the Hx 3 Zone Control Module provide information on the operational status of the zoning system. See Section VII: Hx 3 Zone Control Module - LCD and features for more information about the LCD. The following figure shows the location of the LEDs:

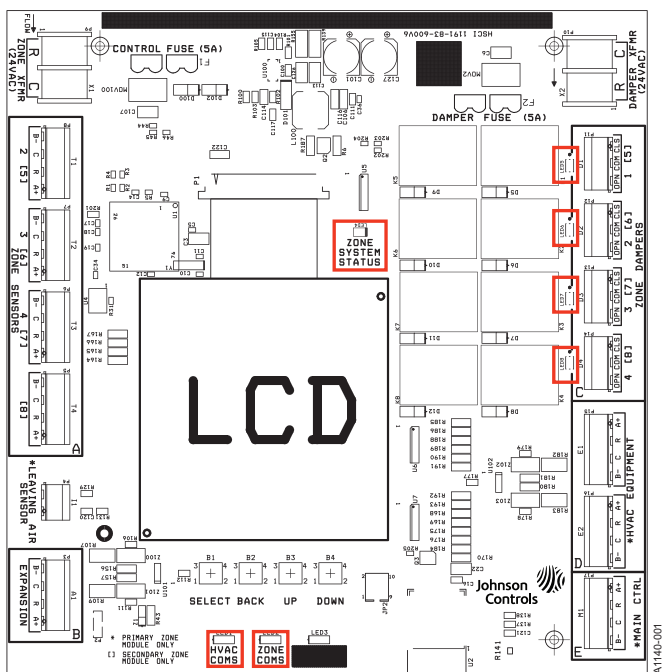


Table 7: Zone system status - LED conditions

LED	Condition
Solid green	Fan is active
Solid blue	Cooling is active
Solid amber	Heating is active
Solid red	Fault is active
Solid yellow	Event is active
Slow green blink	Power applied
Slow yellow blink	LCD ambient out of range (-22°F to 178°F). View the zone status from the Hx 3 Touch Screen Thermostat acting as the Main Control.

Fault and event text is displayed on the Hx 3 Zone Control Module LCD and the appropriate LED color is shown. The following table provides more detailed information about faults and events.

Note: In zoning systems with more than four zones, a second Hx 3 Zone Control Module (Secondary Zone Module) is required for zones 5–8.

Table 8: Hx 3 Zone Control Module - faults and events

Fault or event display text	Zone System Status LED	Flashes	Description	Recommended actions	Display text location	
					Primary Zone Module (zones 1–4)	Secondary Zone Module (zones 5–8)
DAMPER PWR – PHASE – PRI	Red	Solid	Primary Zone Module power and damper power - phase inversion	Measure (voltage) common terminals between Primary Zone Module power and primary damper power. In-phase measures 0 VAC.	X	
DAMPER PWR – PHASE – SEC	Red	Solid	Secondary Zone Module power and damper power - phase inversion	Measure (voltage) common terminals between Secondary Zone Module power and secondary damper power. In-phase measures 0 VAC.	X	X
NO DAMPER PWR - PRIM	Red	Solid	Primary Zone Module - damper power missing	Measure (voltage) R and C terminals - damper power. Voltage must be greater than 24 VAC and less than 32 VAC.	X	
NO DAMPER PWR – SEC	Red	Solid	Secondary Zone Module - damper power missing	Measure (voltage) R and C terminals - damper power. Voltage must be greater than 24 VAC and less than 32 VAC.	X	X
VOLTAGE – BELOW 19 – PRIM	Red	Solid	Primary Zone Module power is less than 19 VAC	Measure (voltage) R and C terminals - Primary Zone Module power. Voltage must be greater than 24 VAC and less than 32 VAC.	X	
VOLTAGE – BELOW 19 – SEC	Red	Solid	Secondary Zone Module power is less than 19 VAC	Measure (voltage) R and C terminals - Secondary Zone Module power. Voltage must be greater than 24 VAC and less than 32 VAC.	X	X
VOLTAGE – BELOW 16 – PRIM	Red	Solid	Primary Zone Module power is less than 16 VAC	Measure (voltage) R and C terminals - Primary Zone Module power. Voltage must be greater than 24 VAC and less than 32 VAC.	X	
VOLTAGE – BELOW 16 – SEC	Red	Solid	Secondary Zone Module power is less than 16 VAC	Measure (voltage) R and C terminals - Secondary Zone Module power. Voltage must be greater than 24 VAC and less than 32 VAC.	X	X
COM ERROR – ZM PRIM	Red	Solid	Loss of communications with Primary Zone Module	Check the four-wire connection (Quadrant B) between the Primary Zone Module and the Secondary Zone Module.		X
				Measure (voltage) R and C terminals - Primary Zone Module power. Voltage must be greater than 24 VAC and less than 32 VAC.		

Table continued on next page

Fault or event display text	Zone System Status LED	Flashes	Description	Recommended actions	Display text location	
					Primary Zone Module (zones 1–4)	Secondary Zone Module (zones 5–8)
COM ERROR – ZM SEC	Red	Solid	Loss of communications with Secondary Zone Module	Check the four-wire connection (Quadrant B) between the Primary Zone Module and the Secondary Zone Module. Measure (voltage) R and C terminals - Secondary Zone Module power. Voltage must be greater than 24 VAC and less than 32 VAC.	X	
COM ERROR – MAIN CTRL	Red	Solid	Loss of communications with the Hx 3 Touch Screen Thermostat acting as the Main Control	Check the four-wire connection (Quadrant E) between the Primary Zone Module and the Main Control. Measure (voltage) R and C terminals - Main Control power. Voltage must be greater than 24 VAC and less than 32 VAC.	X	
COM ERROR – ZONE 2	Red	Solid	Loss of communications with zone 2	Check the four-wire connection (Quadrant A) between the Primary Zone Module and zone 2. Check the zone sensor address configuration - either user interface or DIP switch settings. Measure (voltage) R and C terminals - zone sensor power. Voltage must be greater than 24 VAC and less than 32 VAC.	X	
COM ERROR – ZONE 3	Red	Solid	Loss of communications with zone 3	Check the four-wire connection (Quadrant A) between the Primary Zone Module and zone 3. Check the zone sensor address configuration - either user interface or DIP switch settings. Measure (voltage) R and C terminals - zone sensor power. Voltage must be greater than 24 VAC and less than 32 VAC.	X	
COM ERROR – ZONE 4	Red	Solid	Loss of communications with zone 4	Check the four-wire connection (Quadrant A) between the Primary Zone Module and zone 4. Check the zone sensor address configuration - either user interface or DIP switch settings. Measure (voltage) R and C terminals - zone sensor power. Voltage must be greater than 24 VAC and less than 32 VAC.	X	
COM ERROR – ZONE 5	Red	Solid	Loss of communications with zone 5	Check the four-wire connection (Quadrant A) between the Secondary Zone Module and zone 5. Check the zone sensor address configuration - either user interface or DIP switch settings. Measure (voltage) R and C terminals - zone sensor power. Voltage must be greater than 24 VAC and less than 32 VAC.	X	

Table continued on next page

Fault or event display text	Zone System Status LED	Flashes	Description	Recommended actions	Display text location	
					Primary Zone Module (zones 1–4)	Secondary Zone Module (zones 5–8)
COM ERROR – ZONE 6	Red	Solid	Loss of communications with zone 6	Check the four-wire connection (Quadrant A) between the Secondary Zone Module and zone 6.	X	
				Check the zone sensor address configuration - either user interface or DIP switch settings.		
				Measure (voltage) R and C terminals - zone sensor power. Voltage must be greater than 24 VAC and less than 32 VAC.		
COM ERROR – ZONE 7	Red	Solid	Loss of communications with zone 7	Check the four-wire connection (Quadrant A) between the Secondary Zone Module and zone 7.	X	
				Check the zone sensor address configuration - either user interface or DIP switch settings.		
				Measure (voltage) R and C terminals - zone sensor power. Voltage must be greater than 24 VAC and less than 32 VAC.		
COM ERROR – ZONE 8	Red	Solid	Loss of communications with zone 8	Check the four-wire connection (Quadrant A) between the Secondary Zone Module and zone 8.	X	
				Check the zone sensor address configuration - either user interface or DIP switch settings.		
				Measure (voltage) R and C terminals - zone sensor power. Voltage must be greater than 24 VAC and less than 32 VAC.		
LAS - SHORT	Red	Solid	Leaving air sensor failure (less than 220 Ω). Short condition	Check the two-wire connection for the leaving air sensor input on the Primary Zone Module.	X	
				Remove and measure (resistance) leaving air sensor terminals - Primary Zone Module. Resistance must be greater than 220 Ω .		
LAS - OPEN	Red	Solid	Leaving air sensor failure (greater than 225 k Ω). Open condition	Check the two-wire connection for the leaving air sensor input on the Primary Zone Module.	X	
				Remove and measure (resistance) leaving air sensor terminals - Primary Zone Module. Resistance must be less than 225 k Ω .		
LAS LOW – COOL	Red	Solid	Leaving air sensor is less than installer setting for cool mode	Ensure the leaving air sensor is placed properly - plenum location.	X	
				Check the installer configuration setting. The OEM default is 45°F.		
				Ensure there is proper system refrigeration charge. Check the equipment technical literature.		
				Ensure there is proper airflow across the indoor coil		

Table continued on next page

Fault or event display text	Zone System Status LED	Flashes	Description	Recommended actions	Display text location	
					Primary Zone Module (zones 1–4)	Secondary Zone Module (zones 5–8)
LAS HIGH – HEAT	Red	Solid	Leaving air sensor is greater than installer setting for heat mode	Ensure the leaving air sensor is placed properly - plenum location.	X	
				Check the installer configuration setting. The OEM default is 190°F.		
				Ensure there is proper system refrigeration charge. Check the equipment technical literature.		
				Ensure there is proper airflow across the indoor coil.		
LAS HIGH – HEAT - HP	Red	Solid	Leaving air sensor is greater than installer setting for heat mode - heat pump	Ensure the leaving air sensor is placed properly - plenum location.	X	
				Check the installer configuration setting. The OEM default is 140°F.		
				Ensure there is proper system refrigeration charge. Check the equipment technical literature.		
				Ensure there is proper airflow across the indoor coil.		
ZONE SENSOR FAULT	Red	Solid	Active HVAC equipment communication error or zone sensor fault condition	Active HVAC system fault. Refer to the Main Control.	X	
				Active zone sensor fault - zones 1–8. Refer to the Main Control.		
MISWIRE – ZONE 5	Red	Solid	Zone 5 found without a Secondary Zone Module. Secondary Zone Module is required	Check the four-wire connection (Quadrant B) between the Primary Zone Module and the Secondary Zone Module.	X	
				Remove system power. Install a Secondary Zone Module to Quadrant B.		
				Measure (voltage) R and C terminals - Secondary Zone Module power. Voltage must be greater than 24 VAC and less than 32 VAC.		
MISWIRE – ZONE 5	Red	Solid	Zone 5 found with a Secondary Zone Module	Remove the Main Control from the Secondary Zone Module.	X	X
MISWIRE – INDOOR EQUIP	Red	Solid	Indoor equipment miswire found in Quadrant A or B	Remove system power. Move the indoor equipment four-wire connection to Quadrant D.	X	
MISWIRE – ACCESSORY EQUIP	Red	Solid	Accessory miswire found in Quadrant A or B	Remove system power. Move the accessory four-wire connection to Quadrant D.	X	
MISWIRE – OUTDOOR EQUIP	Red	Solid	Outdoor equipment miswire found in Quadrant A or B	Remove system power. Move the outdoor equipment four-wire connection to Quadrant D.	X	
MISWIRE – ZONE MODULE - SEC	Red	Solid	Secondary Zone Module miswire found in Quadrant D or E	Remove system power. Move the Secondary Zone Module four-wire connection to Quadrant B.	X	
MISWIRE – ZONE THERMOSTAT	Red	Solid	Zone thermostat miswire found in Quadrant D or E	Remove system power. Move the zone thermostat four-wire connection to Quadrant A.	X	
MISWIRE – ZONE SENSOR	Red	Solid	Zone sensor miswire found in Quadrant D or E	Remove system power. Move the zone sensor four-wire connection to Quadrant A.	X	
LEGACY – INDOOR EQUIP	Red	Solid	Legacy indoor equipment miswire found in Quadrant A, B, D, or E	Remove system power. Remove the legacy indoor equipment from the system because it is not supported.	X	

Table continued on next page

Fault or event display text	Zone System Status LED	Flashes	Description	Recommended actions	Display text location	
					Primary Zone Module (zones 1–4)	Secondary Zone Module (zones 5–8)
LEGACY – ACCESSORY EQUIP	Red	Solid	Legacy accessory miswire found in Quadrant A, B, D, or E	Remove system power. Remove the legacy accessory equipment from the system because it is not supported.	X	
LEGACY – OUTDOOR EQUIP	Red	Solid	Legacy outdoor equipment miswire found in Quadrant A, B, D, or E	Remove system power. Remove the legacy outdoor equipment from the system because it is not supported.	X	
LEGACY - ZONE MODULE	Red	Solid	Legacy zone module miswire found in Quadrant A, B, D, or E	Remove system power. Remove the legacy zone module from the system because it is not supported.	X	
LEGACY – ZONE THERMOSTAT	Red	Solid	Legacy zone thermostat miswire found in Quadrant A, B, D, or E	Remove system power. Remove the legacy zone thermostat from the system because it is not supported.	X	
SOFTWARE UPDATE - ACTIVE	Yellow	Solid	Zone software update is active	None	X	
DAMPER TEST - ACTIVE	Green	0.1 s on and 0.1 s off	Damper test mode is active	None	X	
LCD - turned OFF in Software	Yellow	2 s on and 2 s off	Ambient conditions around the Hx 3 Zone Control Module exceed LCD display range	Measure ambient condition at the Hx 3 Zone Control Module location. Ambient temperature or condition is greater than 178°F or less than -22°F. Refer to the Main Control for zone system status information.	X	X

The damper LEDs labeled ZONE DAMPERS indicate the damper position for each zone as shown in Table 9. This position is based on the output signal controlled by the Hx 3 Zone Control Module and does not necessarily reflect the actual position of the zone damper.

Table 9: Zone dampers - LED conditions

LED	Condition
Solid red	Damper is fully closed
Solid green	Damper is fully open
Slow green blink	Damper is partially open
Fast green blink	Damper is opening or closing

The communication LED labeled HVAC COMS indicates active communication on the HVAC communication bus (HVAC equipment and Main Control). The communication LED labeled ZONE COMS indicates active communication on the zone communication bus (Secondary Zone Module and zones 2–8).

Table 10: Communication - LED conditions

LED	Condition
Fast green blink	Communication is active
Off	Communication is not active

Location

Locate the Hx 3 Zone Control Module near the indoor HVAC equipment for convenient wiring of the various zoning system components.

Mounting the Hx 3 Zone Control Module

1. Use the back of the plastic housing to mark four holes on a flat surface near the indoor equipment.
2. Use an 1/8 in. drill bit to drill the four holes where marked in step 1.
3. Use the screws provided to attach the Hx 3 Zone Control Module over the pre-drilled holes.

Wiring

WARNING

ELECTRICAL OPERATION HAZARD

Failure to follow this warning could result in personal injury, death, or equipment damage. Before installing, modifying, or servicing the system, the main electrical disconnect switch must be in the Off position. There may be more than one switch with a suitable warning label.

Wiring requirements

All wiring must comply with local electrical codes and ordinances. Use standard 18 AWG (solid) copper thermostat wires to connect the communicating system. Special (shielded) cable is not generally required. As with all communicating devices, it is best practice to keep wiring at least 1 ft away from large inductive loads. Examples of large inductive loads include electronic air cleaners, high voltage wiring, florescent light ballasts, LED lighting, and motors. If you do not adhere to these wiring practices, electrical interference (noise) may be introduced. Electrical interference may cause erratic system operation.

NOTICE

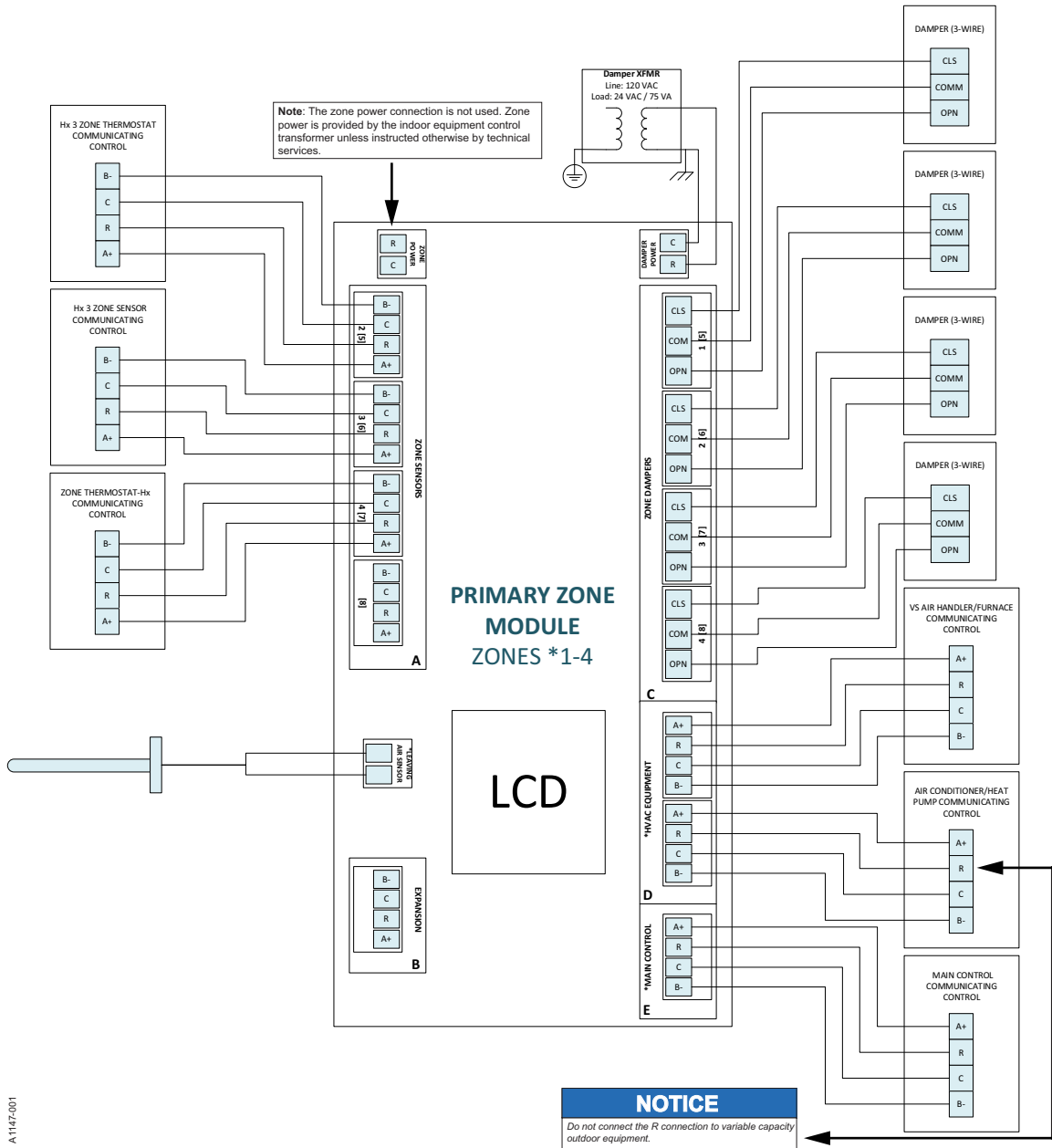
There may be installation applications where large inductive loads cannot be avoided. In these cases, use shielded wire to ensure proper system functionality.

System wiring overview

IMPORTANT: The communicating system requires four wires to operate. Ensure to supply at least four wires to each component of the zoning system.

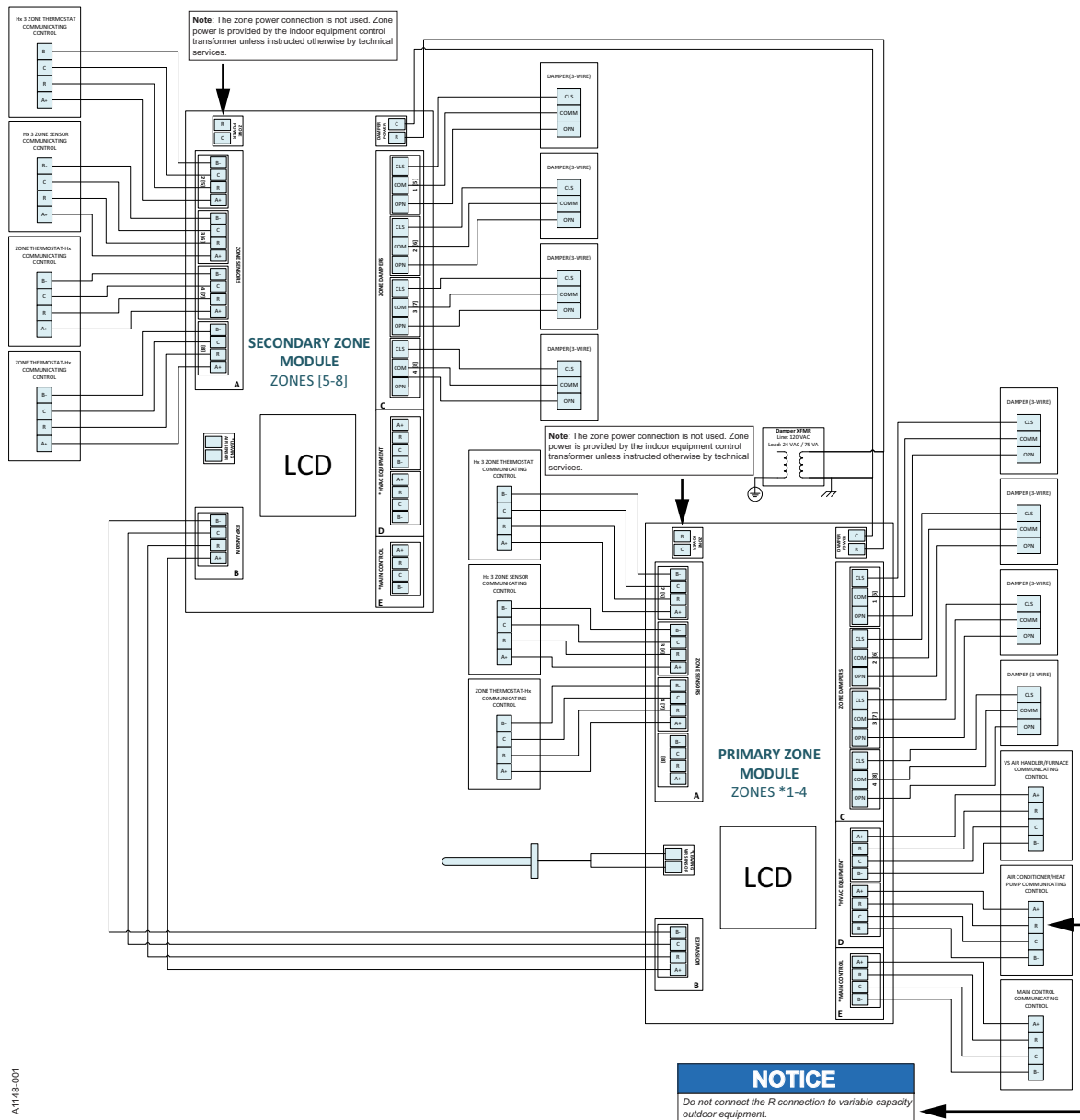
The communicating system is connected by four wires. Two of the wires are used to bring power to the individual zoning components (R and C), and three of the wires are used for serial communication (A+, C, and B-).

Note: For variable capacity outdoor equipment applications, do not connect the indoor 24 VAC R connection to the outdoor equipment.



A1147-001

Figure 8: Power open/power close (three-wire) damper application - example with four zones



A1148-001

Figure 9: Power open/power close (three-wire) damper application - example with eight zones

Main Control wiring (zone 1)

Connect the Hx 3 Touch Screen Thermostat acting as the Main Control to the four-wire screw terminals in Quadrant E (*MAIN CTRL) in the lower right corner of the Primary Zone Module.

Note: See Figure 4 for wiring connections.

NOTICE

If more than one zone thermostat is connected to the zone system, replace the indoor equipment 40 VA transformer with a 75 VA transformer.

Zone sensor wiring (zones 2–8)

Connect the communicating zone sensors to the four-wire screw terminals in Quadrant A (ZONE SENSORS) on the left side of the Hx 3 Zone Control Module. Note the following:

- Use the Primary Zone Module for zones 2–4.
- Use the Secondary Zone Module for zones 5–8 if applicable.

Note: See Figure 5 for wiring connections.

NOTICE

For ease of system configuration and troubleshooting, it is best practice to connect each zone thermostat and/or sensor to the zone number that corresponds to the number assigned to the zone in which it is located. Connect the zone 1 thermostat to the four-wire terminal labeled MAIN CONTROL. Zone thermostats and/or zone sensors (zones 2–8) function properly when connected to any of the four-wire terminals labeled ZONE SENSORS.

Indoor and outdoor equipment wiring

Connect the communicating indoor equipment (furnace or air handler) and communicating outdoor equipment (air conditioner or heat pump) to the Primary Zone Module using the four-wire screw terminals in Quadrant D (*HVAC EQUIPMENT). The indoor equipment transformer provides 24 VAC power to the Hx 3 Zone Control Module and all of the zone sensors, as well as the outdoor equipment and accessory controls.

Variable capacity outdoor equipment does not have an R connection.

Notes: Do not connect the R wire from the *HVAC EQUIPMENT terminal in Quadrant D to variable capacity outdoor equipment.

See Figure 4 for wiring connections.

IMPORTANT: If a modulating furnace is being used as the indoor HVAC equipment, place the ZONE CONTROL jumper in the Yes position. See Figure 10 for the jumper location.

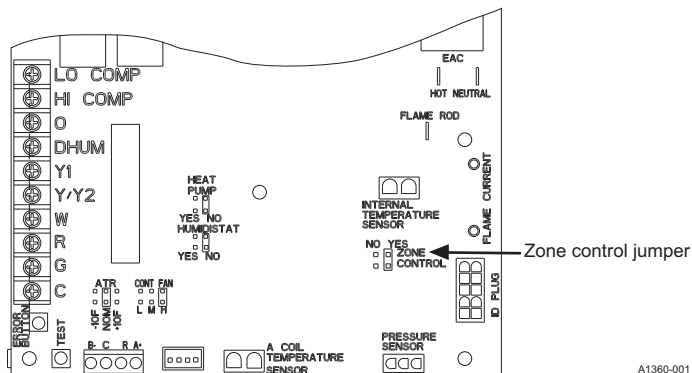


Figure 10: Zone control jumper - modulating furnace

Damper wiring

Connect the zone dampers (purchased separately) to the appropriate three-wire zone damper output screw terminals in Quadrant C (ZONE DAMPERS) of the Hx 3 Zone Control Module. Note the following:

- Use the Primary Zone Module for zones 1–4.
- Use the Secondary Zone Module for zones 5–8 if applicable.

Note: See Figure 4 for wiring connections.

WARNING

Hx 3 ZONE CONTROL MODULE DAMPER WIRE HAZARD

The Hx 3 Zone Control Module is designed to wire a maximum of five (24 VAC) three-wire dampers in parallel, per damper output. Failure to follow this warning could result in personal injury or damage to the Hx 3 Zone Control Module.

NOTICE

Connect the zone dampers to the appropriate zone number as specified before performing the auto setup process on the Main Control.

IMPORTANT: Use consecutive zone numbers, starting with zone 1 as the Main Control. Do not skip zone numbers when identifying the zones and wiring the zone dampers.

Damper power wiring

Connect an external transformer (purchased separately) to the DAMPER XFMR terminals on the upper right side of the Hx 3 Zone Control Module. This external transformer provides power to the zone dampers. A fuse is included in the circuit of the Hx 3 Zone Control Module.

Note: See Figure 4 for wiring connections.

The required power rating for the external transformer depends on the following:

- The number of dampers installed in the system
- The power requirement for each damper

For the damper power requirements, refer to the literature for the dampers being installed. It is important to calculate the actual power required for the dampers selected and size the external transformer appropriately. You can use the transformer sizing table in the Quick Reference Guide included in the Hx 3 Zone Control Module kit to calculate transformer sizing based on the dampers installed. During normal operation, all dampers open or close simultaneously under some conditions, so the rating of the external transformer must meet or exceed the total power requirement for all dampers.

IMPORTANT: Install an external transformer to power the dampers. This transformer is not included with the Hx 3 Zone Control Module kit. Do not connect damper power to the HVAC equipment control transformer.

Leaving air sensor wiring

Connect the wires from the leaving air sensor to the two-wire screw terminals between Quadrant A and Quadrant B (*LEAVING AIR SENSOR) of the Primary Zone Module. Mount the leaving air sensor beyond the indoor coil in the supply air ductwork. Mount the leaving air sensor so it is in the airstream. Extend the leaving air sensor leads as necessary using wire connectors and standard thermostat wire.

Note: See Figure 5 for wiring connections

IMPORTANT: The leaving air sensor must be installed on the Primary Zone Module to allow zoning operation.

Zone expansion wiring (zones 5–8)

If applicable, connect two Hx 3 Zone Control Modules together. To do so, use standard 18 ga. solid thermostat wire to connect the four-wire screw terminals in Quadrant B (EXPANSION) in the lower left corner of each Hx 3 Zone Control Module together.

Note: See Figure 5 for wiring connections.

Section IV: Zoning system installation

The following general procedure covers installation of the zoning system, which includes the Hx 3 Zone Control Module and various other zoning components. Follow the procedure to install and configure the zoning system efficiently and correctly.

Part 1: Prepare for zoning system installation

1. Review the training material and zoning component installation instructions as necessary to familiarize yourself with the features and limitations of the zoning system.
2. Use sound design practices and industry standards to determine the requirements for the equipment, duct system, zone sizes, damper types, and damper sizes. Consider all applicable local codes and ordinances.
3. Determine sensor types, locations, and zone numbers for all zones, based on the homeowner and site requirements.
You must locate the Hx 3 Touch Screen Thermostat (S1-THXU430W) acting as the Main Control in zone 1.
4. Determine the power requirements for the damper motors being used and acquire the appropriate parts.
You can use the Quick Reference Guide included in the Hx 3 Zone Control Module kit to calculate site-specific damper transformer VA requirements. Be aware of the following:
 - Do not connect zone damper power to the HVAC equipment control transformer.
 - An external transformer is required for zone damper power. The Hx 3 Zone Control Module kit does not include a zone damper transformer.

IMPORTANT: Do not connect the damper power connection to the HVAC or zone system transformer.

5. Determine if the indoor transformer needs to be replaced based on zoning system power requirements.
If the zoning system includes zone thermostats in addition to the Hx 3 Touch Screen Thermostat acting as the Main Control, replace the indoor transformer with a 75 VA transformer.
6. Select a location for the Primary Zone Module, the Secondary Zone Module (if applicable), and the leaving air sensor.
See the Location section for more information on selecting a suitable location. Zoning systems with more than four zones require a second Hx 3 Zone Control Module (Secondary Zone Module). You must ensure that the Hx 3 Touch Screen Thermostat acting as the Main Control is wired to the *MAIN CONTROL input on the Primary Zone Module. Failure to do so causes misalignment with zone damper outputs because the Main Control is always zone 1.

IMPORTANT: If there is a second Hx 3 Zone Control Module (Secondary Zone Module) for zones 5-8, ensure the Hx 3 Touch Screen Thermostat acting as the Main Control is wired into the Primary Zone Module. Failure to do causes misalignment with zone damper outputs because the Main Control is always zone 1.

Part 2: Install the zoning system components

1. Complete physical installation of the ductwork, dampers, leaving air sensor, indoor equipment, outdoor equipment, and any system accessories such as ultra-violet lights and electronic air cleaners.
Note the following:
 - Set all control jumpers as required on indoor and outdoor equipment controls.
 - You can adjust the indoor airflow on the Hx 3 Touch Screen Thermostat acting as the Main Control after completing auto setup. To do so, navigate to the **Airflow Configuration** screen from the list of installer settings available.

IMPORTANT: Setting airflow jumpers on the indoor equipment control is required for appropriate airflow ranges. You can use the Main Control to make indoor airflow modifications after completing auto setup. To do so, navigate to the **Airflow Configuration** screen from the list of installer settings available.

2. Update any Hx 3 Touch Screen Thermostats you are using on the zoning system with zoning software if required.
Check the software version on the outside label of the Hx 3 Touch Screen Thermostat box (see Figure 1). When the Hx 3 Touch Screen Thermostat is installed as the Main Control, the software version must be greater than 01.11 to support zoning operation. When the Hx 3 Touch Screen Thermostat is installed as a zone thermostat, the software version must be greater than 02.00 to support zoning operation.
Note: If a Wi-Fi connection is not available at the location where the thermostat is to be installed, apply power to the thermostat. Connect the thermostat to Wi-Fi, and perform a software update before installing the thermostat.
3. Install the Hx 3 Touch Screen Thermostat acting as the Main Control.
4. Install the Primary Zone Module, the Secondary Zone Module (if applicable), and the transformer for zone damper power.
See the Mounting the Hx 3 Zone Control Module section for more information. It is best practice to use a 75 VA transformer at a minimum for zone damper power.
5. Install the zone sensors and configure them with appropriate zone addresses if possible.
The Hx 3 Touch Screen Thermostat acting as the Main Control is always zone 1. You can use the Hx 3 Touch Screen Thermostat (Zone Thermostat-Hx), the Hx 3 Zone Thermostat (Zone Thermostat), and the Hx 3 Zone Sensor (Zone Sensor) as zone sensors. Note the following:
 - You must configure zone thermostats at power-up. Set zone addresses using the touch screen.
 - You can configure Hx 3 Zone Sensors before power-up. Set zone addresses using the DIP switches on the Hx 3 Zone Sensor control board.
6. Wire the zoning components to the Primary Zone Module and the Secondary Zone Module as applicable.
Route wires as follows:
 - Zone thermostats and zone sensors: Quadrant A
 - Zone module expansion: Quadrant B on the Primary Zone Module and the Secondary Zone Module (if applicable). A Secondary Zone Module is required for installations with more than four zones.
 - Leaving air sensor: between Quadrant A and Quadrant B on the Primary Zone Module only
 - Zone dampers: Quadrant C
 - HVAC equipment: Quadrant D on the Primary Zone Module only
 - Main Control (zone 1): Quadrant E on the Primary Zone Module only

Part 3: Complete initial power-up

1. Apply power to the system.
2. Be aware of the following information about software updates to the Hx 3 Touch Screen Thermostat acting as the Main Control:
 - If you did not perform a software update on the Hx 3 Touch Screen Thermostat acting as the Main Control before installation of the zone system and the software version of the thermostat is greater than 01.11, the Hx 3 Zone Control Module updates the thermostat software to its stored version at power-up.
 - If you did not perform a software update on the Hx 3 Touch Screen Thermostat acting as the Main Control before installation of the zone system and the software version of the thermostat is greater than 02.00, the Hx 3 Zone Control Module does not update the thermostat software to its stored version at power-up.
3. Be aware of the following information about software updates to Hx 3 Touch Screen Thermostats acting as zone thermostats:
 - If you did not perform a software update on each Hx 3 Touch Screen Thermostat used as a zone thermostat (Zone Thermostat-Hx) in zones 2–8 before installation and the software version of the thermostats and the Hx 3 Zone Control Module is greater than 02.00, the Hx 3 Zone Control Module will update the thermostat software to its stored version when auto setup is complete.
 - If you did not perform a software update on each Hx 3 Touch Screen Thermostat used as a zone thermostat (Zone Thermostat-Hx) in zones 2–8 before installation and the software version of the thermostats is less than 02.00, the Hx 3 Zone Control Module will not update the thermostat software to its stored version when auto setup is complete.
4. Update the software of Hx 3 Touch Screen Thermostats acting as zone thermostats in zones 2–8 if needed.
 For existing zone thermostats (Zone Thermostat-Hx) being used in zones 2–8:
 - a. Connect each zone thermostat to Wi-Fi individually. Refer to the Installation Manual for the Hx 3 Touch Screen Thermostat for more information about conventional setup.

Or

For new (out of the box) zone thermostats (Zone Thermostat-Hx) being used for zones 2–8, update the thermostats using the Primary Zone Module as follows:

- a. Disconnect the front of the Main Control from the Main Control location on the Primary Zone Module. Do not disconnect the back plate of the thermostat. The Main Control location is in Quadrant E (*MAIN CTRL).
- b. Connect the zone thermostat (Zone Thermostat-Hx) at the Main Control location on the Primary Zone Module.
- c. Apply power to the system.
- d. Wait while the Main Control input on the Primary Zone Module performs a software check on the zone thermostat. If the software version of the zone thermostat is greater than 01.11, the Primary Zone Module updates the thermostat software to its stored version.
- e. When the zone thermostat software update is complete, remove system power.
- f. Disconnect the zone thermostat from the Main Control location on the Primary Zone Module.
- g. Repeat steps b through f for each zone thermostat (Zone Thermostat-Hx) being used for zones 2–8.
- h. When you are finished updating the zone thermostats, ensure the Hx 3 Touch Screen Thermostat acting as the Main Control is connected at the Main Control location on the Primary Zone Module.

Note: The Main Control input on the Primary Zone Module communicates with Hx 3 Touch Screen Thermostats only.

Part 4: Configure equipment and zoning components

1. On the Main Control, on the **Auto Setup** screen, tap **Next** to begin the auto setup process for configuring the indoor and outdoor equipment and zoning components.

IMPORTANT: You must perform initial zone configurations on the Hx 3 Touch Screen Thermostat acting as the Main Control. When you complete the initial zone system profile and finish auto setup, you can reconfigure zone settings on the Main Control or the Primary Zone Module.



A1299-001

Figure 11: Hx 3 Touch Screen Thermostat (Main Control) - Auto setup

2. Troubleshoot component miswires using the Primary Zone Module if required. Otherwise, proceed to step 3.

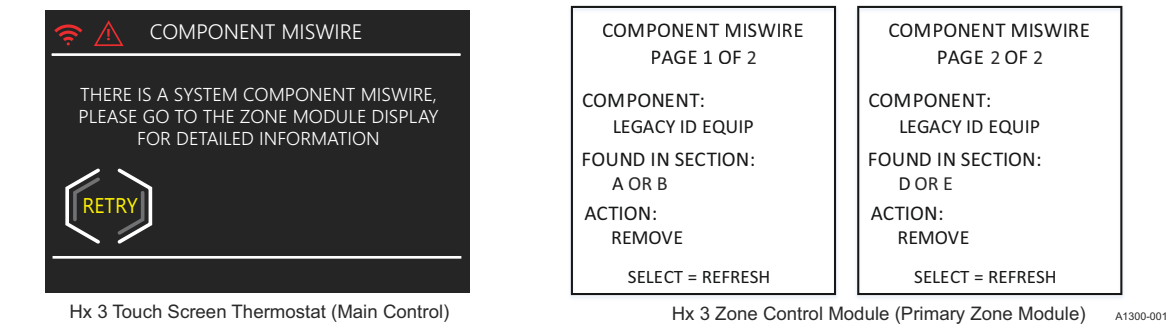


Figure 12: Component miswire on the HVAC or zone Modbus

If component miswires are present on the HVAC or zone Modbus, the Main Control and the Primary Zone Module display a component miswire alert. If this occurs, proceed as follows:

- a. See the Limitations section for a list of supported indoor and outdoor equipment and zone components if required.
- b. If multiple miswires are present, use the **Up** and **Down** buttons on the Primary Zone Module to navigate through the miswires and view the actions required.

**WARNING**

ELECTRICAL MISWIRE HAZARD
Failure to follow this warning could result in personal injury, death, or equipment damage. Before modifying a miswire, ensure the main electrical disconnect switch is in the Off position.

- c. Remove system power because miswires include wires carrying 24 VAC for all communicating controls.
 - d. Perform the required actions to correct the miswires.
 - e. Restore system power.
 - f. On the Main Control, reconfigure the indoor and outdoor equipment through the auto setup process.
3. When prompted, tap **Next** to begin configuring zone thermostats and zone sensors with their appropriate zone addresses.



Figure 13: Zone configuration prompt

4. Configure zone thermostat addresses if required using the **Zone Address** screen as shown in Figure 14.
- Hx 3 Touch Screen Thermostats and/or Hx 3 Zone Thermostats can act as zone thermostats.

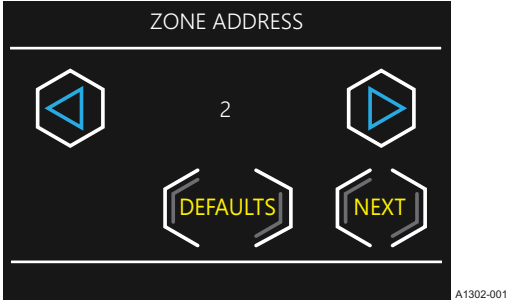


Figure 14: Zone thermostat address configuration - Hx 3 Touch Screen Thermostat (S1-THXU430W) and Hx 3 Zone Thermostat (S1-ZCC01A)

5. Configure Hx 3 Zone Sensor addresses as required using the relevant DIP switches.

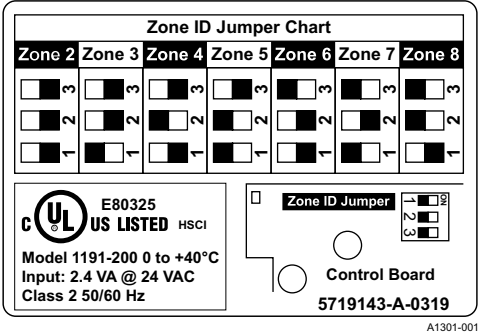


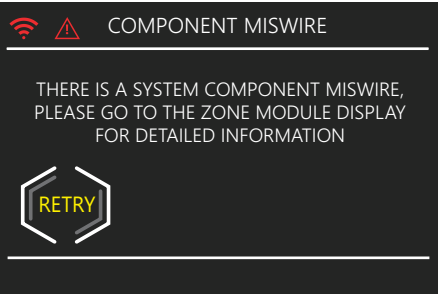
Figure 15: Hx 3 Zone Sensor (S1-TCZSN02) - address configuration

6. Tap **Next**.
The Primary Zone Module searches for different types of zone sensors.

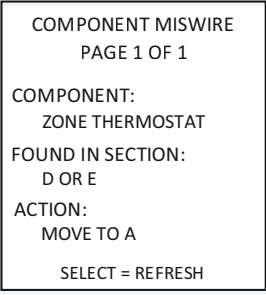


Figure 16: Searching for zone sensors

7. Troubleshoot component miswires using the Primary Zone Module if required. Otherwise, proceed to step 8.



Hx 3 Touch Screen Thermostat (Main Control)



Hx 3 Zone Control Module (Primary Zone Module)

Figure 17: Component miswire - HVAC Modbus

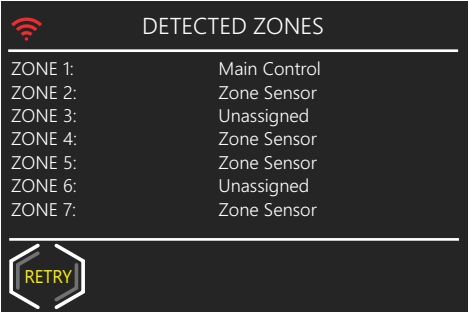
If component miswires are present on the HVAC Modbus, the Main Control and the Primary Zone Module display a component miswire alert. If this occurs, proceed as follows:

- a. See the Limitations section for a list of supported indoor and outdoor equipment and zoning components if required.
- b. If multiple miswires are present, use the **Up** and **Down** buttons on the Primary Zone Module to navigate through the miswires and view the actions required.

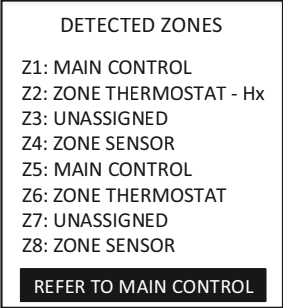
**WARNING**

ELECTRICAL MISWIRE HAZARD
Failure to follow this warning could result in personal injury, death, or equipment damage. Before modifying a miswire, ensure the main electrical disconnect switch is in the Off position.

- c. Remove system power because miswires include wires carrying 24 VAC for all communicating controls.
 - d. Perform the required actions to correct the miswires.
 - e. Restore system power.
 - f. On the Main Control, reconfigure the indoor and outdoor equipment through the auto setup process.
 - g. On the Main Control, rediscover zone sensor types.
8. Troubleshoot unassigned zone sensors if required. Otherwise, proceed to step 9.



Hx 3 Touch Screen Thermostat (Main Control)



Hx 3 Zone Control Module (Primary Zone Module)

Figure 18: Unassigned zones detected

If unassigned zone sensors are present, this is shown on the Main Control and the Primary Zone Module. You must troubleshoot unassigned zone sensors before configuring zone settings. To do so:

- a. Ensure the zone address configurations for all unassigned zones are correct.
 - b. Check the four-wire communication terminals in Quadrant A on the left side of the Primary Zone Module and the Secondary Zone Module (if applicable).
 - c. Check the four-wire communication terminals at the zone locations of all unassigned zone sensors.
 - d. Check the physical four-wire communication wire integrity between the unassigned zone sensor locations and Quadrant A on the left side of the Primary Zone Module and the Secondary Zone Module (if applicable).
 - e. When the unassigned zone configurations and/or wire terminations have been corrected, on the Main Control, tap **Retry**.
 - f. Rediscover zone sensor types using the Main Control.
9. On the **Detected Zones** screen, tap **Next** to begin configuring zone settings.

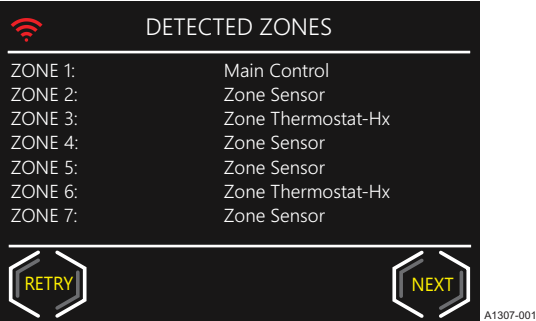


Figure 19: Detected zones

Part 5: Configure settings

1. On the Main Control, on the **Damper Timing** screen, adjust the damper timing for each zone as required. The default setting is 30 s. See Figure 26 for an image of the **Damper Timing** screen on the Main Control. To adjust the damper timing for a zone:
- a. On the **Damper Timing** screen, select the required zone, for example, **Zone 1**.
 - b. Use the **Forward** and **Back** arrows to adjust the damper timing for the zone.
 - c. Tap **Save**.
2. When damper timing settings are complete, tap **Next**.
3. On the **Airflow Bleeding** screen, select the required airflow bleeding settings. Use the scroll bar to view all available settings, and tap the **Yes** or **No** buttons to enable or disable settings. When certain settings are enabled, for example, the setting for allowing airflow to bleed into a dump zone, additional options appear. Tap the **Yes** or **No** buttons and the **Forward** and **Back** arrows to enable, disable, or adjust these settings as required. The following table provides an overview of the airflow bleeding settings. See the Airflow bleeding section for an image of the airflow bleeding settings (Figure 27) on the Main Control and a detailed explanation of each airflow bleeding option.

Table 11: Airflow bleeding settings

Name	Default setting	Available settings	Additional options when setting is enabled
Allow Airflow to Bleed into a Dump Zone?	No	Yes or No	Allow airflow to bleed into a dump zone in cooling mode. The default setting is Yes. Tap Yes or No . Specify a cool dump zone. The default setting is zone 2. Allow airflow to bleed into a dump zone in heating mode. The default setting is Yes. Tap Yes or No . Specify a heat dump zone. The default setting is zone 2.
Allow Airflow Bleeding into Zones in Same Mode?	Yes	Yes or No	
Allow Airflow Bleeding into Zones in Off Mode?	Yes	Yes or No	
Allow Airflow Bleeding into Zones in Opposite Mode?	Yes	Yes or No	

4. When airflow bleeding settings are complete, tap **Next**.
5. On the **Zone Safeties** screen, adjust the leaving air safety settings for cooling, heating, and heating - hp (heat pump) as required.
- To adjust a setting:
- a. On the **Zone Safeties** screen, select the required setting, for example, **LAS - Cooling**.
 - b. Use the **Forward** and **Back** arrows to adjust the temperature.
 - c. Tap **Save**.
- See Figure 40 for an image of the **Zone Safeties** screen on the Main Control and an example of a screen for adjusting an individual zone safety setting. Figure 40 shows the **LAS - Cooling** screen. The **LAS - Heating** and **LAS - Heating - HP** screens have the same options but the default temperature settings are different. The following table gives an overview of the leaving air safety settings.

Table 12: Leaving air safety settings

Name	Default setting
LAS - Cooling	45°F
LAS - Heating	190°F
LAS - Heating - HP	140°F

6. When leaving air safety settings are complete, tap **Next**.
7. On the **Zone Airflow** screen, adjust the airflow for each zone as required.

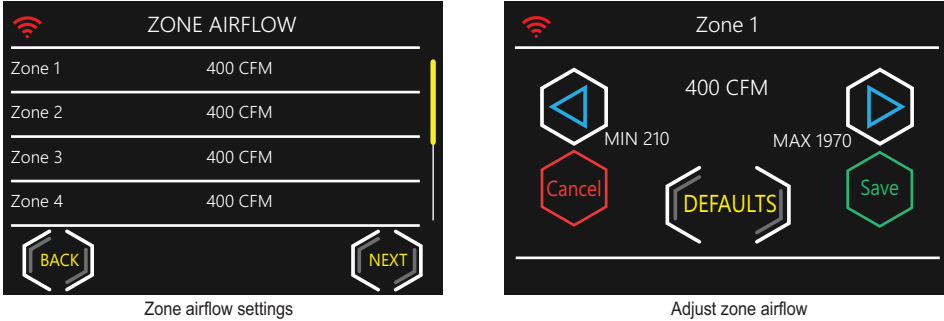


Figure 20: Zone airflow

The default setting for each zone is 400 CFM. Determine the maximum airflow requirements for each zone.

Note: If you have opted to use the OEM default settings for bleeding excess airflow (step 3) or opted to bleed excess airflow with a single setting, apply the required OEM 60% airflow multiplier per zone. Add the multiplier value to the site-specific load calculated value per zone. Use the Quick Reference Guide included in the Hx 3 Zone Control Module kit to calculate site-specific maximum airflow requirements.

IMPORTANT: If you have opted to use the OEM default settings for bleeding excess airflow or opted to bleed excess airflow with a single setting, apply the required OEM 60% airflow multiplier per zone.

- To adjust the airflow for a zone:
- a. On the **Zone Airflow** screen, select the required zone, for example, **Zone 1**.
 - b. Use the **Forward** and **Back** arrows to adjust the airflow for the zone. Note that the maximum value displayed beside the **Forward** arrow and the minimum value displayed beside the **Back** arrow are the maximum and minimum CFM that the indoor blower can deliver on the low and high side of the system per model.
 - c. Tap **Save**.
8. When zone airflow settings are complete, tap **Next**.
9. On the **Zone Names** screen, change the name of each zone as required.

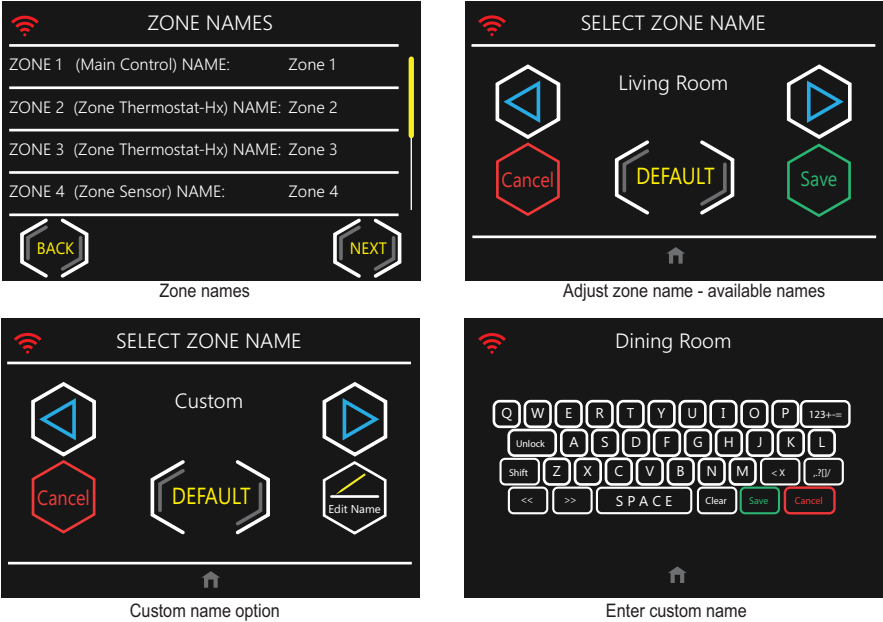


Figure 21: Zone names

- The default name for each zone is the zone number. For example, zone 1 is named Zone 1 and zone 2 is named Zone 2. To change the name of a zone:
- On the **Zone Names** screen, select the zone.
 - On the **Select Zone Name** screen, use the **Forward** and **Back** arrows to navigate through the available names. Select the required name, for example, Living Room, and tap **Save**.
- Alternatively, create a custom zone name with a maximum of 15 characters. To do so, select the **Custom** option, tap **Edit Name**, use the keyboard to enter the required zone name, for example, Dining Room, tap **Save**, then tap **Save** again.
10. When zone name changes are complete, tap **Next**.
11. Review the **Equipment Summary** screen to confirm the correct number of zone sensors has been detected.

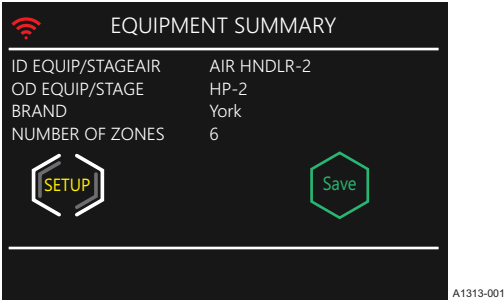


Figure 22: Equipment summary

The **Equipment Summary** screen displays when the initial zone system profile is complete. The number of zones shown reflects the number of zone sensors the Primary Zone Module detected during the auto setup process. This is the same information as shown on the **Detected Zones** screen. Reviewing the **Equipment Summary** screen provides another opportunity to confirm the correct number of zone sensors has been detected.

- 12. Tap **Save**.
- 13. On the **System Settings** screen, configure additional zoning system settings (installer settings) as required.

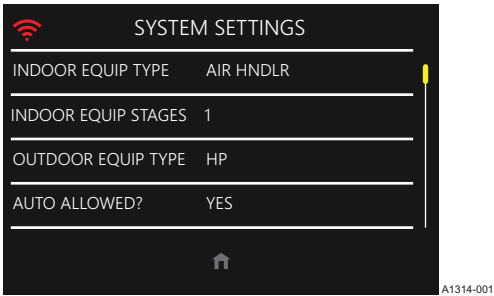


Figure 23: System settings

Part 6: Instruct the homeowner

- 1. Inform the homeowner about the zoning system features and operation.
- 2. Refer the homeowner to the User's Information Manual for more information.

Section V: Zoning system settings

The Hx 3 Touch Screen Thermostat acting as the Main Control and the Hx 3 Zone Control Module acting as the Primary Zone Module have various common settings. These settings are for use by installers. See the following for detailed information on zoning system settings.

Accessing zone settings

To access zone settings, go to the **Zoning Menu** screen on the Main Control or the **Zone Settings** screen on the LCD on the Primary Zone Module. See Figure 24.

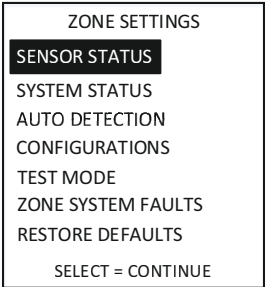
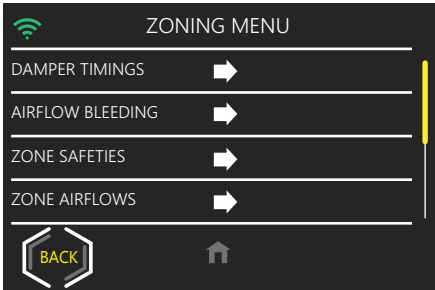


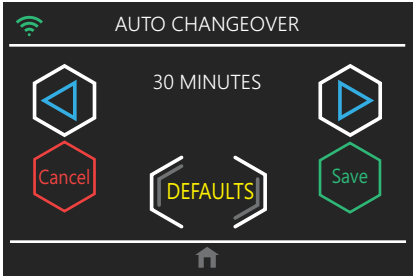
Figure 24: Zone settings

Heat/cool changeover time

Go to the **Auto Changeover** screen on the Main Control or the Primary Zone Module to adjust the heat/cool changeover time as required. See Figure 25. The heat/cool changeover time is the amount of time that elapses before the Primary Zone Module does the following:

- Energizes the system for cooling if the system is currently in heating
- Energizes the system for heating if the system is currently in cooling

This setting applies when different zones in the system have simultaneous calls for heating and cooling.



Hx 3 Touch Screen Thermostat (Main Control)



Hx 3 Zone Control Module (Primary Zone Module)

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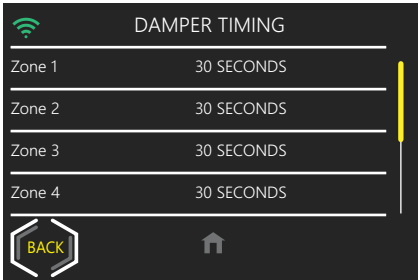
Figure 25: Auto changeover

Damper timing

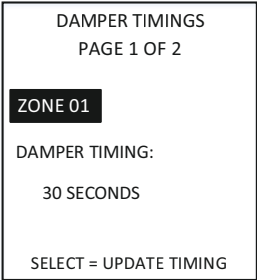
Go to the **Damper Timing** screen on the Main Control or the **Damper Timings** screen on the Primary Zone Module to adjust damper timing as required. See Figure 26. The damper time is the time required to fully open or fully close the damper. Obtain this value from the technical specifications for the damper being used.

NOTICE

Actual damper times may vary from specified values. For the best results, measure the actual time required to fully open or fully close the damper being installed in each zone, and use this value on the screen for damper timing.



Hx 3 Touch Screen Thermostat (Main Control)



Hx 3 Zone Control Module (Primary Zone Module)

A1317-001

Figure 26: Damper timing

IMPORTANT: Accurate damper time settings are important for proper system operation.

Types of dampers

The Hx 3 Zone Control Module is designed to wire a maximum of five (24 VAC) power open/power close three-wire dampers in parallel, per damper output. Failure to adhere to this could result in personal injury or damage to the Hx 3 Zone Control Module.

IMPORTANT: The Hx 3 Zone Control Module supports 24 VAC power open/power close three-wire dampers only. Two-wire dampers are not supported.

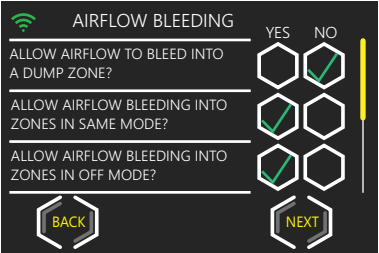
WARNING

ZONE MODULE DAMPER WIRE HAZARD

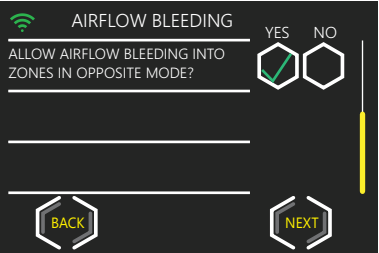
The Hx 3 Zone Control Module is designed to wire a maximum of five (24 VAC) three-wire dampers in parallel, per damper output. Failure to follow this warning could result in personal injury or damage to the Hx 3 Zone Control Module.

Airflow bleeding

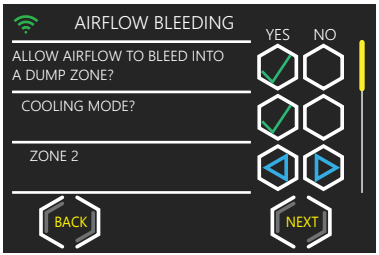
The zoning system provides options for managing airflow restrictions that are present for most zoning installations. Airflow restrictions occur when the amount of airflow required for system operation exceeds the airflow required for calling zones (that is, zone demand). Go to the **Air-flow Bleeding** screen on the Main Control or the Primary Zone Module to adjust airflow bleeding settings as required. See Figure 27.



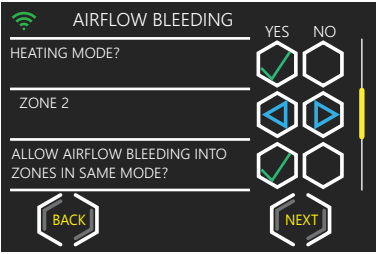
Hx 3 Touch Screen Thermostat (Main Control)



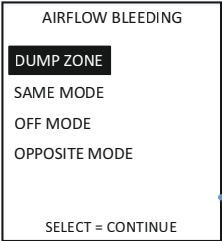
Hx 3 Touch Screen Thermostat (Main Control)



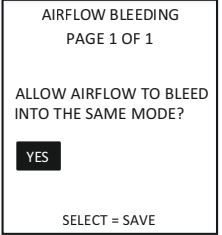
Hx 3 Touch Screen Thermostat (Main Control)



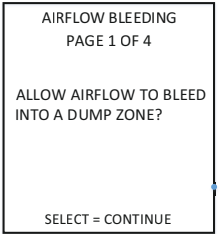
Hx 3 Touch Screen Thermostat (Main Control)



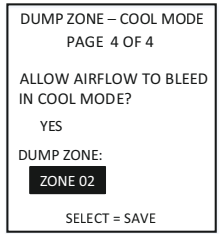
Hx 3 Zone Control Module (Primary Zone Module)



Hx 3 Zone Control Module (Primary Zone Module)



Hx 3 Zone Control Module (Primary Zone Module)



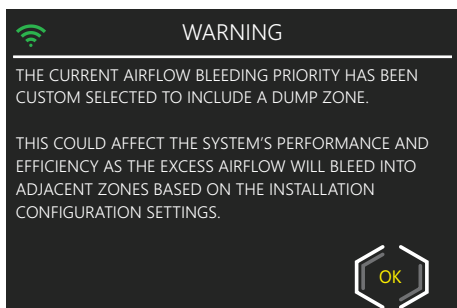
Hx 3 Zone Control Module (Primary Zone Module)

A1318-001

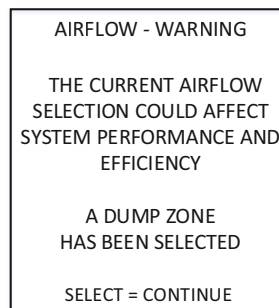
Figure 27: Airflow bleeding

The following options are available for airflow bleeding:

- Allow airflow bleeding into a dump zone: Allows you to select a dump zone for either heating or cooling modes of operation. The Hx 3 Zone Control Module partially opens the dump zone damper based on the airflow required for proper operation. If the Hx 3 Zone Control Module determines the dump zone damper needs to be fully open to manage the excess airflow, the Hx 3 Zone Control Module fully opens the dump zone damper. If you select a dump zone, the Main Control or the Primary Control Module displays a warning as shown in the following figure:



Hx 3 Touch Screen Thermostat (Main Control)



Hx 3 Zone Control Module (Primary Zone Module)

A1319-001

Figure 28: Dump zone warning

- Allow airflow bleeding into zones in same mode: Allows the Hx 3 Zone Control Module to partially open dampers to zones that are operating in the same mode (heating or cooling) and do not have a call for conditioning. This option is always used when airflow bleeding is enabled.
- Allow airflow bleeding into zones in off mode: Allows the Hx 3 Zone Control Module to partially open dampers to zones that are in off mode (that is, not in heating or cooling mode) and do not have a call for conditioning.
- Allow airflow bleeding into zones in opposite mode: Allows the Hx 3 Zone Control Module to partially open dampers to zones that are in opposite mode (that is, heating when in cooling mode or cooling when in heating mode) and do not have a call for conditioning.

Zoning system installations with a bypass damper (not required)

NOTICE

The Hx 3 Zone Control Module is designed to manage excess airflow. A bypass damper is not required.

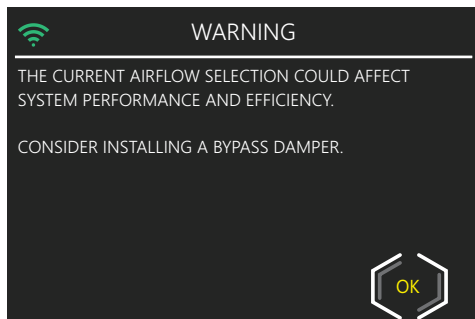
In typical zoning installations, the primary option to manage airflow restrictions is the use of a bypass damper. The Hx 3 Zone Control Module is designed to manage excess airflow. A bypass damper is not required.

Note the following information:

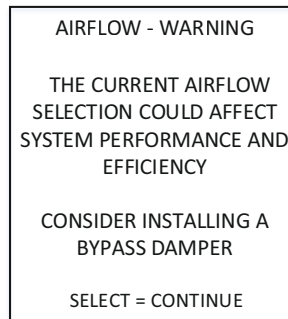
- If a bypass damper is installed, enable or disable the setting to allow excess airflow bleeding into other zones by setting it to **Yes** or **No**.
- If you have opted to use the OEM default settings for bleeding excess airflow for zones in same, off, or opposite mode or opted to bleed excess airflow with a single setting, apply the required OEM 60% airflow multiplier per zone. Add the multiplier value to the site-specific load calculated value per zone. Use the Quick Reference Guide included in the Hx 3 Zone Control Module kit to calculate site-specific maximum airflow requirements.

IMPORTANT: If you have opted to use the OEM default settings for bleeding excess airflow or opted to bleed excess airflow with a single setting, apply the required OEM 60% airflow multiplier per zone.

- If the maximum airflow of an individual zone is set lower than the minimum airflow of the indoor equipment, or if excess airflow is allowed to bleed into zones in same, off, or opposite mode, the Main Control or the Primary Zone Module displays a warning as shown in Figure 29.



Hx 3 Touch Screen Thermostat (Main Control)



Hx 3 Zone Control Module (Primary Zone Module)

A1320-001

Figure 29: Bypass damper warning

- If using a bypass damper, it must be designed as part of the duct system. Do not connect the bypass damper to the Hx 3 Zone Control Module. This means the Hx 3 Zone Control Module does not control or monitor the bypass damper operation. A three-wire 24 VAC power damper can be used as a bypass damper and controlled by external duct static pressure using S1-SPS switch kit. An S1-SPRD barometric bypass damper can be used as an alternative to a powered damper.

Zoning system installations without a bypass damper

The preferred option for managing airflow restrictions is the airflow bleeding functionality of the zoning system. With this functionality, the Hx 3 Zone Control Module redirects excess airflow to other zones by partially opening dampers to non-calling zones when the airflow required for proper equipment operation exceeds the airflow required by calling zones.

Be aware of the following information:

- If a bypass damper is not installed and airflow restrictions are anticipated, enable the setting to allow excess airflow bleeding into other zones by setting it to **Yes**.
- If you have opted to use the OEM default settings for bleeding excess airflow for zones in same, off, or opposite mode or opted to bleed excess airflow with a single setting, apply the required OEM 60% airflow multiplier per zone. Add the multiplier value to the site-specific load calculated value per zone. Use the Quick Reference Guide included in the Hx 3 Zone Control Module kit to calculate site-specific maximum airflow requirements.

IMPORTANT: If you have opted to use the OEM default settings for bleeding excess airflow or opted to bleed excess airflow with a single setting, apply the required OEM 60% airflow multiplier per zone.

- When the system is operating, the Hx 3 Zone Control Module continually determines if there is a need to bleed excess airflow to other zones. This determination is based on the airflow required for proper operation at the time, considering equipment limitations, the stage of operation and other factors, and the airflow settings for each zone calling for conditioning. If the Hx 3 Zone Control Module determines there is a need to implement the airflow bleeding functionality, it uses each of the enabled settings to manage the airflow to the various zones, while maintaining the setpoint of each zone as closely as possible. The enabled settings are implemented in the order shown on the **Airflow Bleeding** screen (from the top down).

Maximum airflow for individual zones

IMPORTANT: Determine the number of zones and the airflow for each of these zones by a room by room load calculation. You can then select HVAC equipment, and proceed to implementing a proper duct design. Proper system design and installation is critical for correct operation of the zoning system.

Determine airflow settings for each zone based on a room by room load calculation and the required CFM per zone. The duct system must be properly designed. Proper system design and installation is critical for correct operation of the zoning system.

Set the CFM for individual zones to a minimum of 100 CFM or the maximum allowable airflow of the equipment (indoor model dependent). The default setting is 400 CFM.

If the maximum airflow setting of the zone is less than the minimum airflow of the indoor equipment, the Hx 3 Zone Control Module bleeds excess airflow into zones that do not have a call for conditioning.

The zone airflow settings allow the configuration and evaluation of maximum airflow for individual zones. Go to the **Zone Airflow** screen on the Main Control or the **Zone Airflows** screen on the Primary Zone Module to adjust zone airflow settings as required. The following figure shows the zone airflow setting for an individual zone:

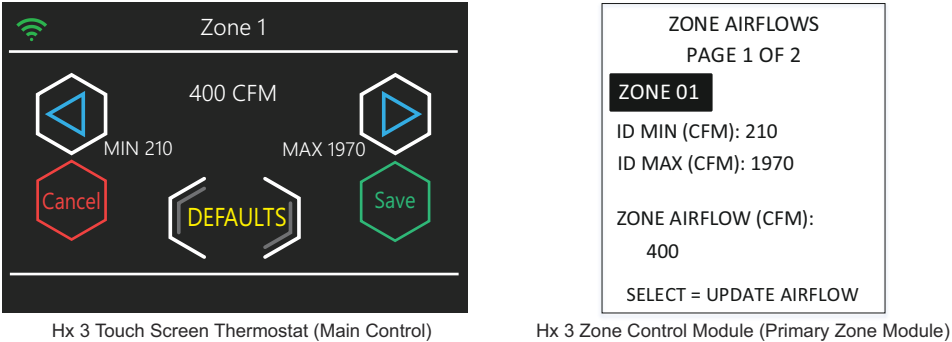


Figure 30: Maximum airflow for individual zone - setting screen

NOTICE

The airflow values on the setting screens for zone airflow are the airflows the Hx 3 Zone Control Module commands the indoor unit to deliver. The actual airflow delivered to each zone may differ from these airflow values for some installations.

Airflow warnings

If the total airflow specified for all zones exceeds the maximum airflow that can be delivered by the indoor unit, the Main Control or the Primary Zone Module displays a warning as shown in Figure 31. You can reduce the airflow to some or all zones, depending on the installation conditions.

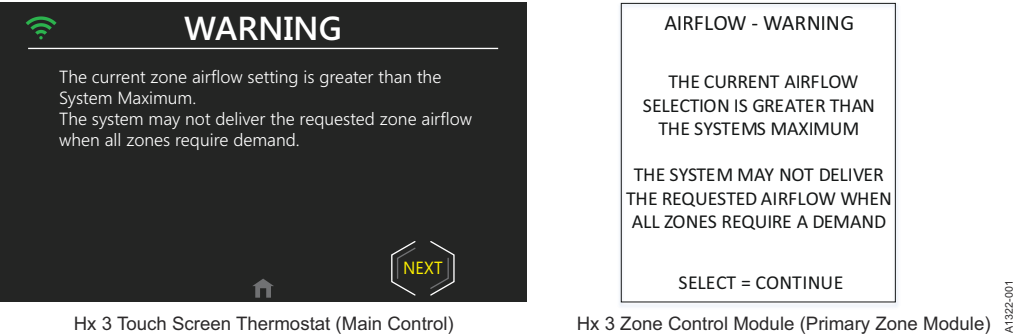


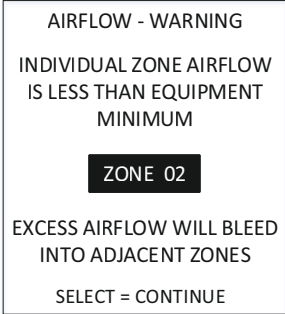
Figure 31: Warning - airflow setting greater than system maximum

If the airflow for a zone is less than the minimum airflow that can be delivered by the indoor unit, the Main Control or the Primary Zone Module displays a warning as shown in Figure 32. The warning shows the affected zone and indicates that the airflow bleeding functionality will apply. Depending on the installation conditions, do one of the following:

- Increase the airflow to some or all zones
- Reconfigure the zone layout of the site



Hx 3 Touch Screen Thermostat (Main Control)



Hx 3 Zone Control Module (Primary Zone Module)

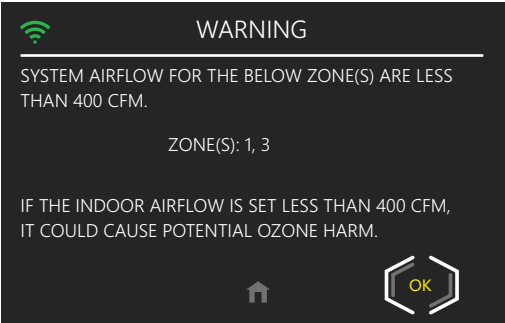
A1323-001

Figure 32: Warning - zone airflow less than equipment minimum

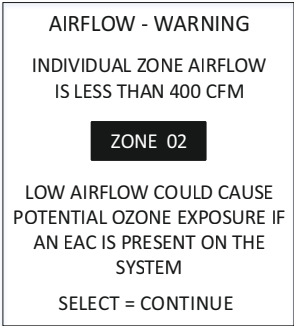
If the airflow for a zone is less than 400 CFM, the Main Control or the Primary Zone Module displays a warning as shown in Figure 33. The warning shows the affected zones and indicates ozone harm could potentially occur.

Depending on the installation conditions, do one of the following:

- Increase the airflow to some or all zones
- Reconfigure the zone layout of the site



Hx 3 Touch Screen Thermostat (Main Control)



Hx 3 Zone Control Module (Primary Zone Module)

A1324-001

Figure 33: Warning - zone airflow less than 400 CFM

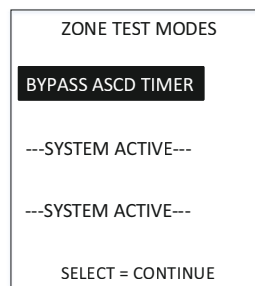
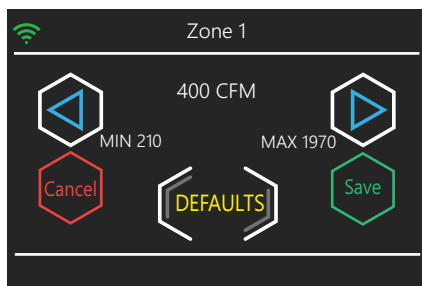
Forced airflow testing for an individual zone

Note: Completion of the initial zone system profile through the auto setup process on the Main Control is required before testing individual zone airflows. You cannot perform a forced zone airflow test if the zone system profile is not configured, or if there is an active system demand (for example, cooling, heat, or fan only), or if damper test mode is in use.

The zoning system allows airflow testing for individual zones. You can perform a forced airflow test for a zone when the following conditions are met:

- An active system demand (for example, cooling, heat, or fan only) is not present
- Damper test mode is not in use
- The zone system profile has been configured

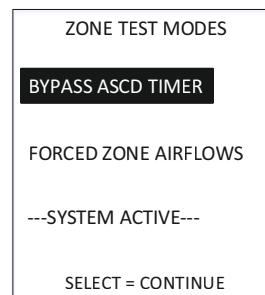
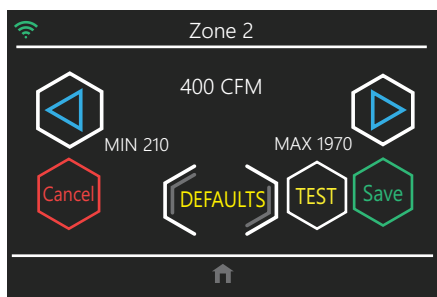
Figure 34 shows the Main Control and the Primary Zone Module when the option to perform a forced airflow test is not available. Figure 35 shows the Main Control and the Primary Zone Module when the option to perform a forced airflow test is available.



Hx 3 Touch Screen Thermostat (Main Control or Zone Thermostat-Hx)

Hx 3 Zone Control Module (Primary Zone Module)

A1325-001

Figure 34: Forced airflow test is not available because there is an active system demand

Hx 3 Touch Screen Thermostat (Main Control or Zone Thermostat-Hx)

Hx 3 Zone Control Module (Primary Zone Module)

A1326-001

Figure 35: Forced airflow test is available

NOTICE

You can only test one zone during a forced airflow test. To test a different zone, perform another forced airflow test.

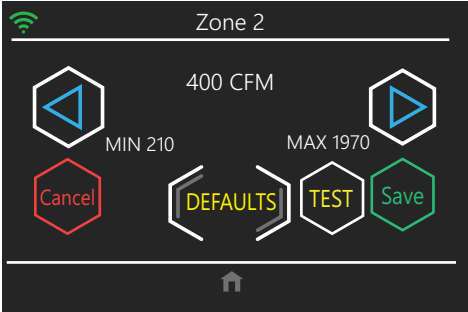
You can enable forced airflow tests for individual zones from various zoning components as follows:

- Hx 3 Touch Screen Thermostat acting as the Main Control: Can be used to perform forced airflow tests for all zones in the system (up to eight zones).
- Hx 3 Touch Screen Thermostat acting as a zone thermostat (Zone Thermostat-Hx): Can be used to perform a forced airflow test for its individual zone.
- Hx™ Thermostat Pro App: Can be used to perform forced airflow tests for all zones in the system (up to eight zones). A Wi-Fi connection and approval from the homeowner (granted through the Hx™ Thermostat App for homeowners) are required to use this option.
- Hx 3 Zone Control Module acting as Primary Zone Module: Can be used to perform forced airflow tests for all zones in the system (up to eight zones).

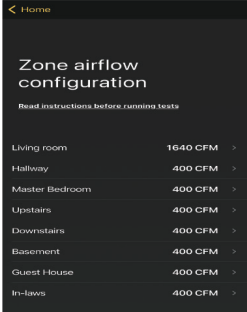
The preferred option to manage airflow restrictions is the airflow bleeding functionality of the zoning system. With this functionality, the Hx 3 Zone Control Module redirects excess airflow to other zones by partially opening dampers to non-calling zones when the airflow required for proper equipment operation exceeds the airflow required by calling zones. If airflow bleeding is not allowed, install a bypass damper to manage airflow restrictions.

If the maximum airflow setting of a zone is less than the minimum airflow (CFM) of the indoor equipment, the Hx 3 Zone Control Module redirects excess airflow to adjacent zones. You can view the following on the screens of the different zoning components (see Figure 36):

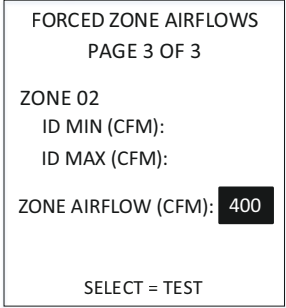
- The minimum and maximum CFM of the indoor equipment
- The CFM settings of the individual zone



Hx 3 Touch Screen Thermostat
(Main Control or Zone Thermostat-Hx)



Hx Thermostat Pro App



Hx 3 Zone Control Module
(Primary Zone Module)

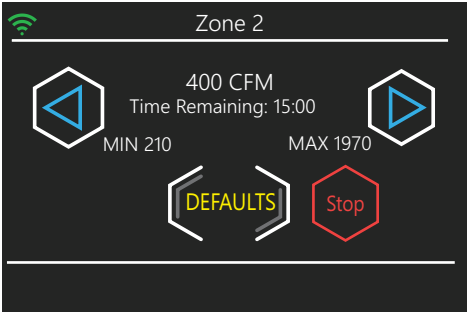
Figure 36: Individual zone - maximum airflow testing

NOTICE

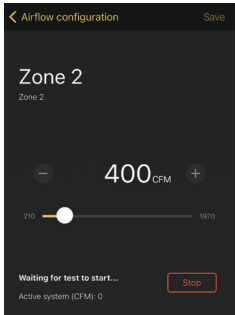
*When you select the **Test** button, the Hx 3 Zone Control Module closes all dampers except the dampers for the zone being tested. The indoor unit does not deliver airflow to the zone being tested until the dampers are in their test positions. Expect a delay that is at least as long as the greatest damper time specified for the application.*

It is important to be aware of the following when performing a forced airflow test:

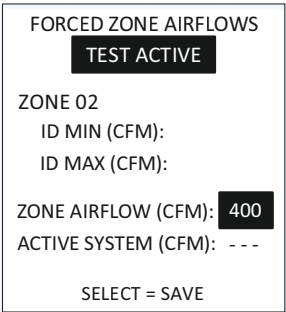
- You can increase or decrease the active CFM by 10 CFM at a time during a forced airflow test for a zone if you enable the test from one of the following: Hx 3 Touch Screen Thermostat (Main Control), Hx 3 Touch Screen Thermostat (Zone Thermostat-Hx), or Hx 3 Zone Control Module (Primary Zone Module).
- You cannot adjust the active CFM during a forced airflow test for a zone if you enable the test from the Hx Thermostat Pro App. In this case, you can only test a constant CFM.
- If a forced airflow test is in progress and the indoor blower has not communicated its active CFM (meaning the requested CFM has not been delivered to the zone being tested yet), the active system CFM is not displayed on the screens of the different zoning components, as shown in the following figure:



Hx 3 Touch Screen Thermostat
(Main Control or Zone Thermostat-Hx)



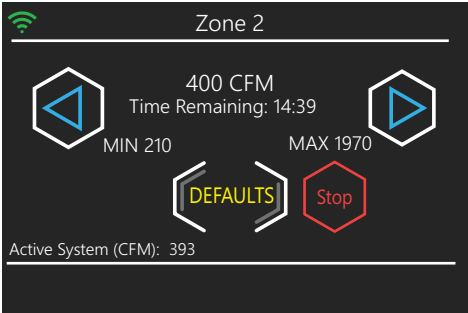
Hx Thermostat Pro App



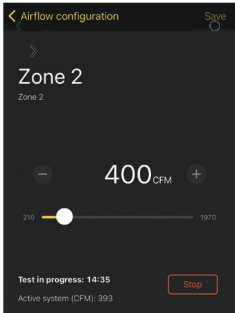
Hx 3 Zone Control Module
(Primary Zone Module)

Figure 37: Active test screen

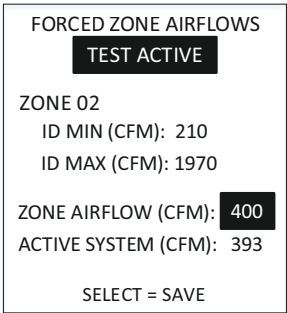
- If a forced airflow test is in progress and the indoor blower has communicated its active CFM (meaning the requested CFM has been delivered to the zone being tested), the active system CFM is displayed on the screens of the different zoning components, as shown in Figure 38.



Hx 3 Touch Screen Thermostat
(Main Control or Zone Thermostat-Hx)



Hx Thermostat Pro App



Hx 3 Zone Control Module
(Primary Zone Module)

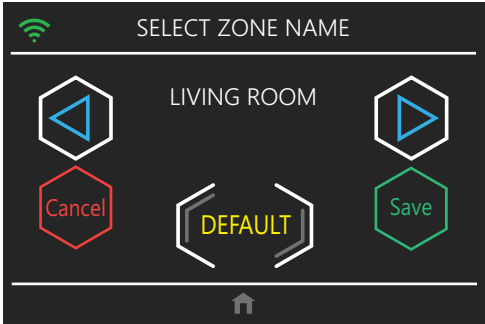
A1329-002

Figure 38: Active test screen

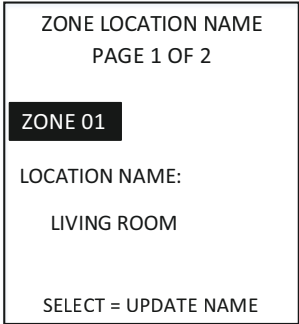
- You can tap the **Defaults** button on the Hx 3 Touch Screen Thermostat (Main Control or Zone Thermostat-Hx) to change the airflow of the zone to 400 CFM.
- To save the airflow value being tested, tap the **Save** button or press the **Select** button, depending on which zoning component screen you are using. If you do not want to save the airflow value being tested, tap the **Stop** button or press the **Back** button, depending on which zoning component screen you are using.

Zone names

The default system name for each zone is the zone number. Use the **Zone Names** screen on the Main Control or the **Zone Location Name** screen on the Primary Zone Module to change the names of individual zones. See Figure 39. Choose names from the list of stored names on the system as required. On the Main Control, you also have the option to create custom names with a maximum of 15 characters. The homeowner can change zone names at any time.



Hx 3 Touch Screen Thermostat (Main Control)



Hx 3 Zone Control Module (Primary Zone Module)

A1330-001

Figure 39: Zone names

Note: You cannot create a custom name for a zone using the Primary Zone Module. Use the Hx 3 Touch Screen Thermostat (Main Control or Zone Thermostat-Hx) or the Hx Thermostat app (for homeowners) to create custom names.

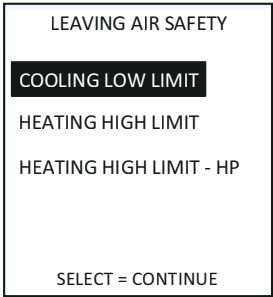
NOTICE
Custom zone names are limited to a maximum of 15 characters.

Zone safeties

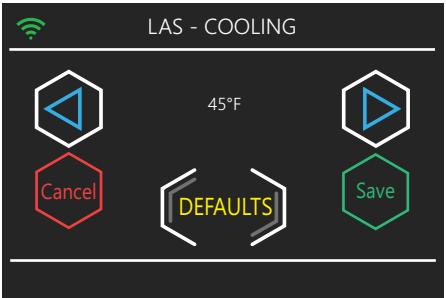
Use the **Zone Safeties** screen on the Main Control or the **Leaving Air Safety** screen on the Primary Zone Module to specify the minimum leaving air temperature allowed for cooling and the maximum leaving air temperature allowed for heating. See Figure 40. These limits are based on the temperature measured by the leaving air sensor and are only used for protection of the system. The leaving air sensor value is not used in the comfort control algorithm.



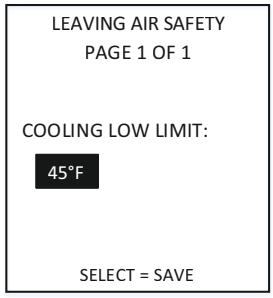
Hx 3 Touch Screen Thermostat (Main Control)



Hx 3 Zone Control Module (Primary Zone Module)



Hx 3 Touch Screen Thermostat (Main Control)



Hx 3 Zone Control Module (Primary Zone Module)

A1331-001

Figure 40: Leaving air safety

NOTICE

The installation location of the leaving air sensor and the ductwork configuration are unique for each site. The zone safety settings may need to be adjusted for some configurations. The leaving air sensor must be installed in a location before the location of any zone dampers.

Section VI: Zoning system software updates

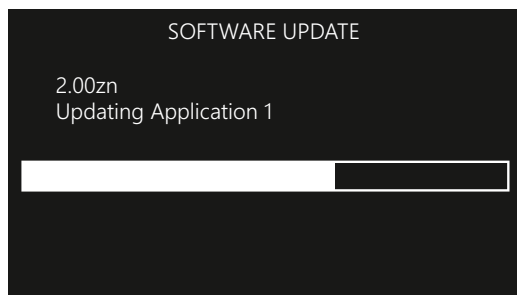
The following zoning components indicate when software updates are in progress:

- Hx 3 Touch Screen Thermostat acting as the Main Control
- Zone Thermostats - Hx 3 Touch Screen Thermostat (Zone Thermostat-Hx) and Hx 3 Zone Thermostat (Zone Thermostat)
- Hx 3 Zone Control Module - Primary Zone Module and Secondary Zone Module

When software updates are complete, the components display the screen shown before the software update, for example, the **Home** screen.

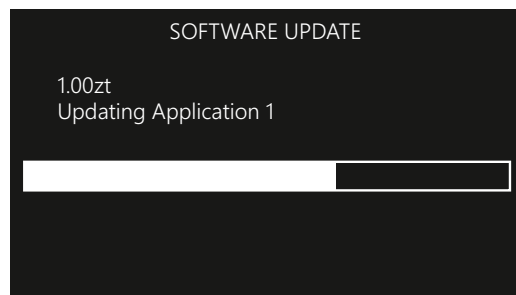
Main Control and zone thermostats

When receiving a software update, the Main Control and zone thermostats display a black background with a white progress bar in the middle of the screen, as shown in the following figure:



Hx 3 Touch Screen Thermostat (Main Control) - zone 1
or

Hx 3 Touch Screen Thermostat (Zone Thermostat-Hx) - zones 2–8

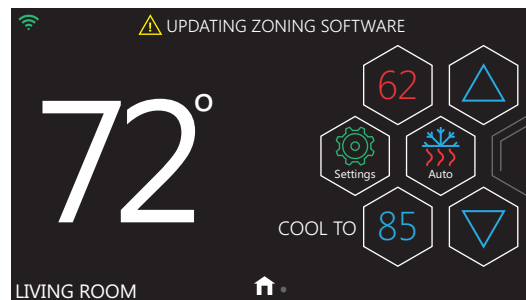


Hx 3 Zone Thermostat (Zone Thermostat) - zones 2–8

A1332-001

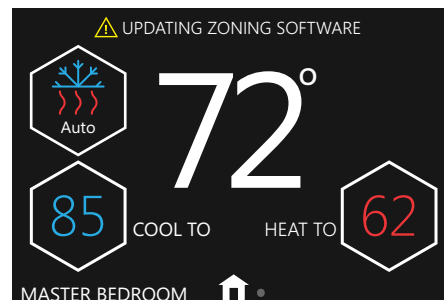
Figure 41: Main Control and zone thermostat - receiving a software update

When the Main Control and zone thermostats are not receiving a software update, but an update to a zoning component is in progress, a yellow triangle and text appear in their banner area, as shown in the following figure:



Hx 3 Touch Screen Thermostat (Main Control) - zone 1
or

Hx 3 Touch Screen Thermostat (Zone Thermostat-Hx) - zones 2–8



Hx 3 Zone Thermostat (Zone Thermostat) - zones 2–8

A1333-001

Figure 42: Main Control and zone thermostat - software update banner indicator

Hx 3 Zone Control Module

When the Primary Zone Module is receiving a software update from the Main Control, the following occur:

- The Zone System Status LED on the Primary Zone Module shows as green, and a progress bar is displayed in the middle of the screen.
- The Zone System Status LED on the Secondary Zone Module shows as yellow and the **General Display** screen is shown.

See Figure 43.

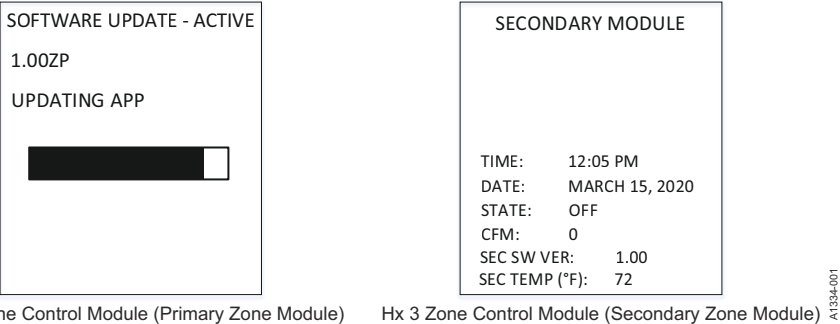


Figure 43: Primary Zone Module receiving a software update

When the Secondary Zone Module is receiving a software update from the Primary Zone Module, the following occur:

- The Zone System Status LED on the Secondary Zone Module shows as green, and a progress bar is displayed in the middle of the screen.
- The Zone System Status LED on the Primary Zone Module shows as yellow and two progress bars are displayed on the screen. The progress bar in the middle of the screen shows the progress of the Secondary Zone Module software update. The overall progress bar at the bottom of the screen shows the progress of overall zoning component software updates.

See Figure 44.

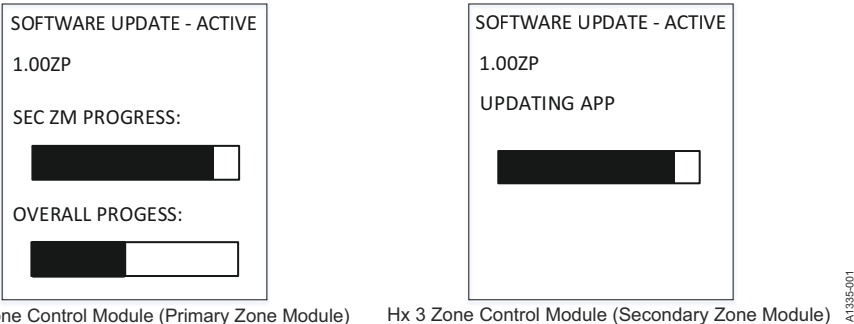
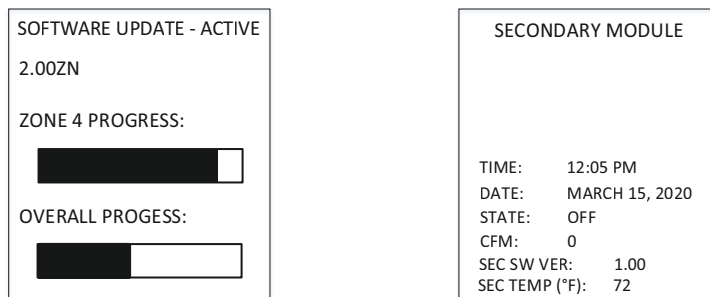


Figure 44: Primary Zone Module sending software update to Secondary Zone Module

When the Primary Zone Module is sending a software update to a zoning component, the following occur:

- The Zone System Status LED on the Primary Zone Module shows as yellow and two progress bars are displayed on the screen. The progress bar in the middle of the screen shows the progress of the zoning component software update. The overall progress bar at the bottom of the screen shows the progress of overall zoning component software updates.
- The Zone System Status LED on the Secondary Zone Module shows as yellow and the **General Display** screen is shown.

See Figure 45.



Hx 3 Zone Control Module (Primary Zone Module) Hx 3 Zone Control Module (Secondary Zone Module)

A1336-001

Figure 45: Primary Zone Module sending software update to zoning component

Section VII: Hx 3 Zone Control Module - LCD and features

The Hx 3 Zone Control Module can act as the Primary Zone Module (zones 1–4) or the Secondary Zone Module (zones 5–8) on the zoning system. Zoning systems with more than four zones require a Secondary Zone Module for zones 5–8. The Primary Zone Module has functionality and settings that are not available on the Secondary Zone Module. The following provides detailed information about using the Hx 3 Zone Control Module LCD and the features available on the Primary Zone Module and the Secondary Zone Module.

Note: Completion of the initial zone system profile through the auto setup process on the Main Control is required before you can access the **Zone Settings** screen (installer settings) on the Hx 3 Zone Control Module.

Using the LCD

The following are some useful points to note when using the LCD on the Hx 3 Zone Control Module:

- Press the **Select**, **Back**, **Up**, or **Down** buttons on the Hx 3 Zone Control Module to access the **General Display** screen.
- Press and hold the **Up** and **Down** buttons at the same time for more than 5 s to access the **Zone Settings** screen. See Figure 51. The zone settings are for use by installers.
- Press the **Up** and **Down** buttons to navigate the menu.
- Use the **Select** button as indicated on specific screens, for example, to save settings or discover zone sensors.
- Press the **Back** button to return to the previous screen.
- If a page count is shown on a screen, this indicates the number of additional pages (that is, screens) available for view.
- Certain settings and test modes are not available when there is a system demand for cooling or heating, or when forced zone airflow tests and damper tests are in progress. If a setting or test mode is not available, **System Active** is displayed on the screen in place of the setting or test mode name.
- To reduce energy usage, the LCD turns off after five min if you do not interact with Hx 3 Zone Control Module screens. The only exception to this is the **Zone Settings** screen. The LCD turns off after 120 min if you do not interact with the **Zone Settings** screen.

Incomplete profile

Complete the initial zone system profile through the auto setup process on the Main Control before accessing the **Zone Settings** screen (installer settings) on the Hx 3 Zone Control Module. The **Profile Incomplete** screen displays on the Primary Zone Module and the Secondary Zone Module until the zone system profile is configured on the Main Control.

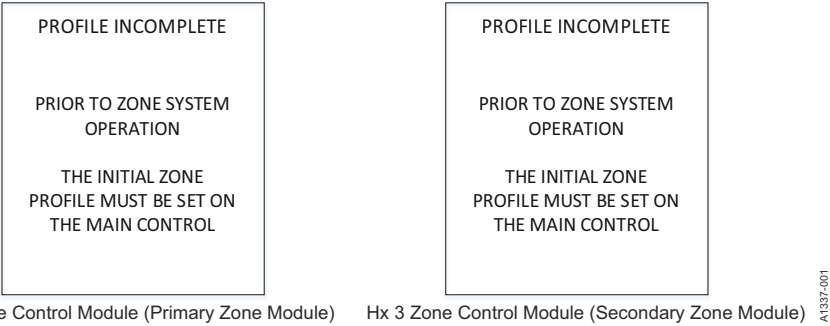


Figure 46: Incomplete profile

Unassigned zone sensor

Refer to the Main Control if an unassigned zone sensor is found during the initial installation. See Figure 47.

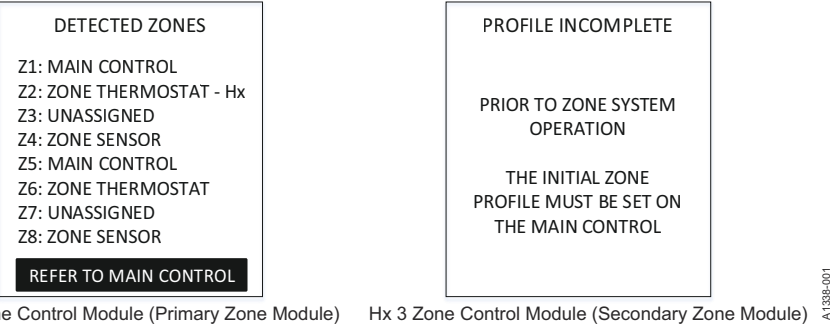
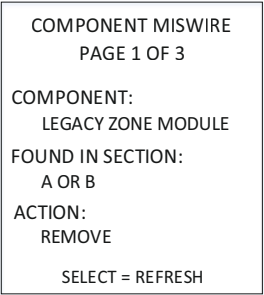


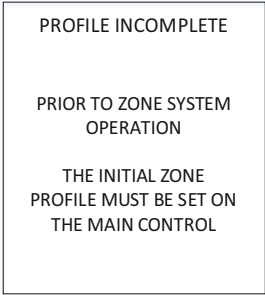
Figure 47: Unassigned zone sensor

Component miswire

If a component miswire is present during initial system configuration, the **Component Miswire** screen displays on the Primary Zone Module. See Figure 48. Perform the action shown on the **Component Miswire** screen. In the example shown in Figure 48, the action is to remove a legacy zone module from the system. When the component miswire is corrected, press **Select** on the Primary Zone Module to clear the miswire fault.



Hx 3 Zone Control Module (Primary Zone Module)



Hx 3 Zone Control Module (Secondary Zone Module)

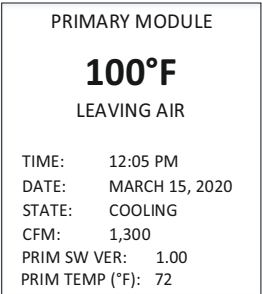
A1389-001

Figure 48: Component miswire

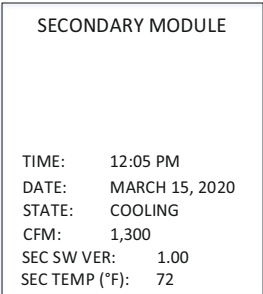
General display

Press the **Select**, **Back**, **Up**, or **Down** buttons on the Hx 3 Zone Control Module to access the **General Display** screen. This screen is slightly different on the Primary Zone Module and the Secondary Zone Module. See Figure 49.

Note: If active faults are present, the **Active Fault** screen is displayed instead of the **General Display** screen.



Hx 3 Zone Control Module (Primary Zone Module)



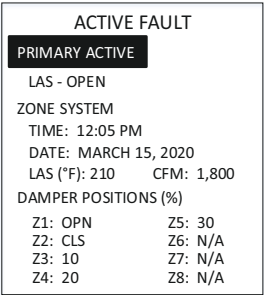
Hx 3 Zone Control Module (Secondary Zone Module)

A1340-001

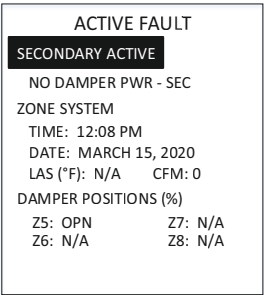
Figure 49: General display

Active faults

If active faults are present and you press the **Select**, **Back**, **Up**, or **Down** buttons on the Hx 3 Zone Control Module, the **Active Fault** screen is displayed. The **Active Fault** screen shows the active fault with the highest priority. See Figure 50.



Hx 3 Zone Control Module (Primary Zone Module)



Hx 3 Zone Control Module (Secondary Zone Module)

A1341-001

Figure 50: Active fault

Software updates

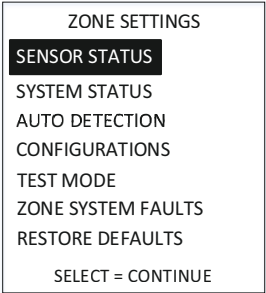
When the Hx 3 Zone Control Module is receiving a software update, a screen showing the progress of the update is displayed. See Figure 43 for an image of the Primary Zone Module receiving a software update. See Figure 44 for an image of the Secondary Zone Module receiving a software update. You cannot access the **Zone Settings** screen during a software update.

When the Primary Zone Module is sending a software update to a zoning component, a screen showing the progress of the update is displayed. The Secondary Zone Module shows the **General Display** screen at this time. See Figure 45. See Section VI: Zoning system software updates for more information about software updates.

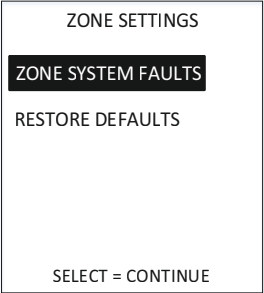
Zone settings

Press and hold the **Up** and **Down** buttons on the Hx 3 Zone Control Module at the same time for more than 5 s to access the **Zone Settings** screen. The zone settings are for use by installers. The Primary Zone Module has more available settings than the Secondary Zone Module. See Figure 51 and the detailed explanations of the zone settings that follow for more information.

Note: The LCD turns off after 120 min if you do not interact with the **Zone Settings** screen.



Hx 3 Zone Control Module (Primary Zone Module)



Hx 3 Zone Control Module (Secondary Zone Module)

A1343-001

Figure 51: Zone settings (installer settings)

Sensor status - Primary Zone Module only

Zone sensor status information is view only. You can only view the **Zone Sensor Status** screen on the Primary Zone Module. See Figure 52. The number of pages available for view reflects the number of zones configured on the zoning system.

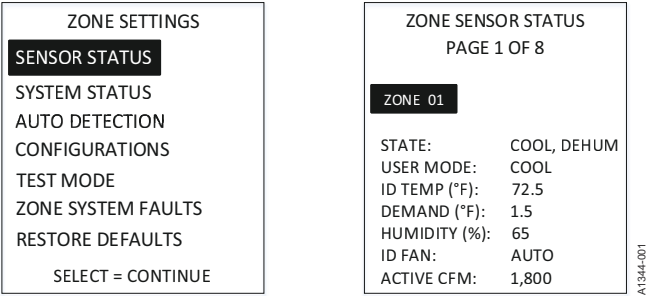


Figure 52: Zone sensor status

System status - Primary Zone Module only

Zone system status information is view only. You can only view the **Zone System Status** screen on the Primary Zone Module. The **Zone Sensor Status** screen shows HVAC and zone system information. See Figure 53.

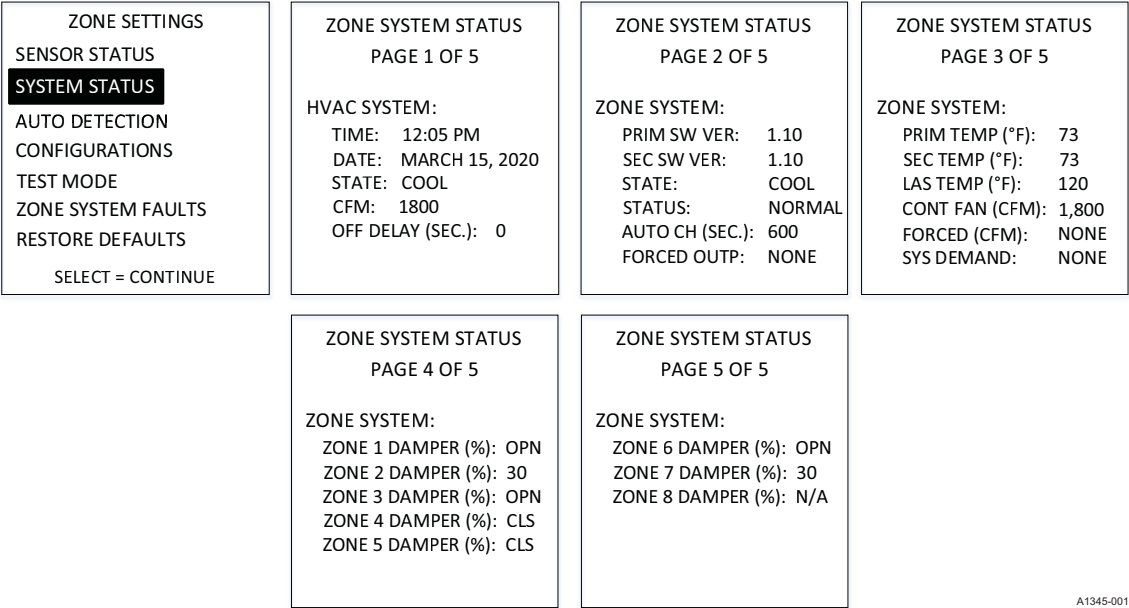


Figure 53: Zone system status

Auto detection - Primary Zone Module only

Zone auto detection information is view only. You can only view the **Auto Detection** screen on the Primary Zone Module. The **Auto Detection** screen shows HVAC airflows and detected zones on the zone system. In the example shown in Figure 54, you would press **Select** on page 2 or page 3 (that is, screen 2 or screen 3) to enable zone sensor discovery.

The Primary Zone Module identifies the type and number of zone sensors installed on the zoning system during initial system configuration. If a new type of zone sensor is added to the zone system or the number of zone sensors changes at any time, you do not have to use the Main Control for zone sensor discovery. Use the **Discover** option on the Primary Zone Module to determine the new zone sensor types and the number of zone sensors. You do not need to use the **Restore Defaults** option.

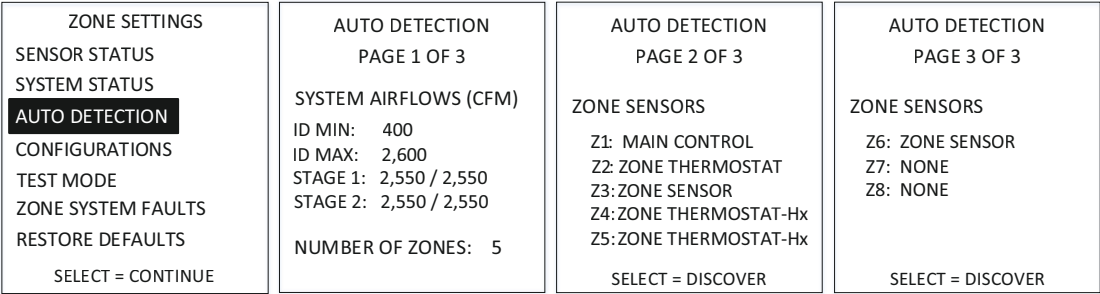


Figure 54: Zone auto detection

Configurations - Primary Zone Module only

Use the Main Control or the Primary Zone Module to access zone configurations settings, for example, airflow bleeding and leaving air safety settings.

If there is a system demand for cooling or heating or if a forced zone airflow test or damper test is in progress, certain settings and test modes are not available. If a setting or test mode is not available, **System Active** is displayed on the screen in place of the setting or test mode name. In the example of the **Zone Configurations** screen on the Primary Zone Module in Figure 55, **System Active** is displayed where the leaving air safety setting appears when available.

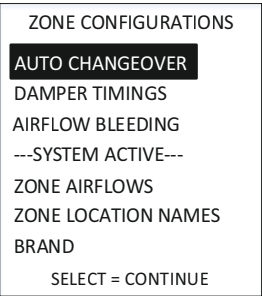


Figure 55: Setting unavailable because there is an active system demand

Note: You cannot create a custom name for a zone using the **Zone Location Names** option on the Primary Zone Module.

NOTICE

Custom zone names can only be created using the Hx 3 Touch Screen Thermostat (Main Control or Zone Thermostat-Hx) or the Hx Thermostat app.

Test mode - Primary Zone Module only

Select the **Test Mode** option to access the zone test modes available. Use the **Zone Test Modes** screen to do the following:

- Bypass the HVAC system timers
- Test and adjust airflows for individual zones
- Perform manual or automated testing of damper output timings

All zones must be in off mode before you can perform a forced zone airflow test or a damper test. When there is an active system demand or another test mode is in use, test modes are not available. **System Active** is displayed on the screen in place of the test mode name. See the examples in Figures 56 and 57.

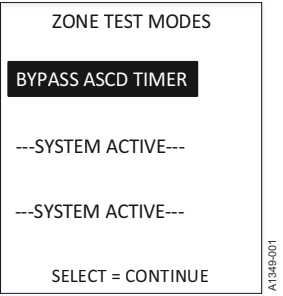


Figure 56: Damper test and forced zone airflow test unavailable because there is an active system demand

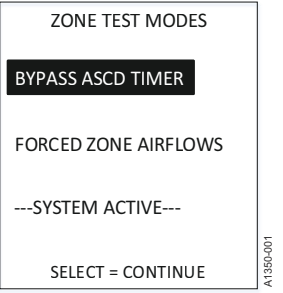


Figure 57: Damper test unavailable because there is an active system demand or forced zone airflow test in progress

NOTICE

The forced zone airflows and damper test modes are available on the Primary Zone Module when all zones are in off mode.

Bypassing the anti-short cycle delay (ASCD) timer - Primary Zone Module only

1. On the **Zone Test Modes** screen, use the **Up** and **Down** buttons to navigate to the **Bypass ASCD timer** option and press **Select**.
2. On the **ASCD Timer** screen, press **Select** to confirm you want to bypass the timer.

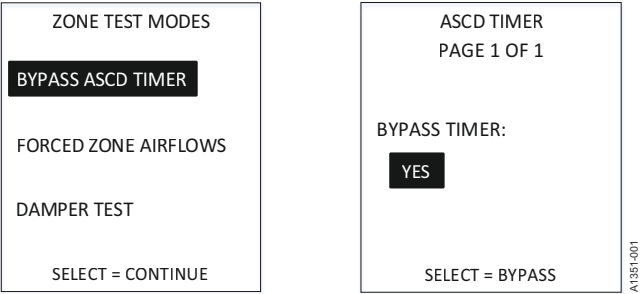


Figure 58: Bypass ASCD timer

Performing a forced airflow test - Primary Zone Module only

1. On the **Zone Test Modes** screen, use the **Up** and **Down** buttons to navigate to the **Forced Zone Airflows** option and press **Select**.
2. On the **Forced Zone Airflows** screen, use the **Up** and **Down** buttons to change the zone if required and navigate to the page (or screen) showing the test option.

You have the option to update airflow CFM on the **Forced Zone Airflows** screen. To do so, press **Select** to confirm you want to update the airflow CFM, use the **Up** and **Down** buttons to update the airflow CFM, and press **Select** to save the change.

3. Press **Select** to confirm you want to perform the test.
While the test is running, you can press **Back** at any time to end the test.
4. Press the **Up** or **Down** buttons to increase or decrease the active CFM by 10 CFM at a time if required.
5. Press **Select** to save the airflow CFM setting being tested or press **Back** if you do not want to save the airflow CFM setting being tested.

ZONE TEST MODES BYPASS ASCD TIMER FORCED ZONE AIRFLOWS DAMPER TEST SELECT = CONTINUE	FORCED ZONE AIRFLOWS PAGE 1 OF 3 ZONE 01 ID MIN (CFM): ID MAX (CFM): ZONE AIRFLOW (CFM): 400 SELECT = UPDATE AIRFLOW	FORCED ZONE AIRFLOWS PAGE 2 OF 3 ZONE 01 ID MIN (CFM): ID MAX (CFM): ZONE AIRFLOW (CFM): 1400 SELECT = SAVE	FORCED ZONE AIRFLOWS PAGE 3 OF 3 ZONE 01 ID MIN (CFM): ID MAX (CFM): ZONE AIRFLOW (CFM): 1400 SELECT = TEST
	FORCED ZONE AIRFLOWS TEST ACTIVE ZONE 01 ID MIN (CFM): ID MAX (CFM): ZONE AIRFLOW (CFM): 1400 ACTIVE SYSTEM (CFM): - - - SELECT = SAVE	FORCED ZONE AIRFLOWS TEST ACTIVE ZONE 01 ID MIN (CFM): 210 ID MAX (CFM): 1970 ZONE AIRFLOW (CFM): 1400 ACTIVE SYSTEM (CFM): 1404 SELECT = SAVE	

A1352-001

Figure 59: Forced zone airflow test

Performing a damper test - Primary Zone Module only

You can perform two types of damper test:

- Automated damper test: Used to test each zone in the zoning system in sequence
- Manual damper test: Used to test a specific zone in the zoning system

To perform an automated damper test:

1. On the **Zone Test Modes** screen, use the **Up** and **Down** buttons to navigate to the **Damper Test** option and press **Select**.
2. On the **Damper Test Screen**, use the **Up** and **Down** buttons to navigate to the **Automated** option and press **Select**.
3. On the **Automated Test** screen, use the **Up** and **Down** buttons to change the damper output time and test time if required.
4. Press **Select** to confirm you want to start the test.

While the test is running, you can press **Back** at any time to end the test.

During an automated damper test, the following occur:

- The damper LED for zone 1 shows as green and the output remains fully open for the duration of the configured test time. Damper voltage is only delivered to the normally open damper contacts for the duration of the configured damper timing per zone.
- The damper LEDs for the other zones in the zoning system (zones 2–8) emit green flashes until all the dampers have fully closed. The timing is based on the configured damper timing per zone. Damper voltage is only delivered to the normally closed damper contacts for the duration of the configured damper timing per zone.
- When the zone 1 test is complete, the zone system begins testing zone 2. Each zone in the zoning system is tested in sequence until all zones have been tested.
- When the automated damper test is complete, the zoning system enters standby mode.

To perform a manual damper test:

1. On the **Zone Test Modes** screen, use the **Up** and **Down** buttons to navigate to the **Damper Test** option and press **Select**.
2. On the **Damper Test Screen**, use the **Up** and **Down** buttons to navigate to the **Manual** option and press **Select**.
3. On the **Manual Test** screen, use the **Up** and **Down** buttons to change the damper output time and the zone damper if required and press **Select**.
4. Press **Select** to confirm you want to start the test.

While the test is running, you can press **Back** at any time to end the test.

During a manual damper test, the following occur:

- The damper LED for the zone being tested shows as green and the output remains fully open for the duration of the configured test time. Damper voltage is only delivered to the normally open damper contacts for the duration of the configured damper timing per zone.
- The zone damper LEDs for the other zones in the zoning system (the zones not selected for testing) emit green flashes until all the dampers have fully closed. The timing is based on the configured damper timing per zone. Damper voltage is only delivered to the normally closed damper contacts for the duration of the configured damper timing per zone.
- When the manual damper test is complete, the zoning system enters standby mode.

ZONE TEST MODES BYPASS ASCD TIMER FORCED ZONE AIRFLOWS DAMPER TEST SELECT = CONTINUE	DAMPER TEST TEST TYPE: AUTOMATED MANUAL SELECT = CONTINUE	AUTOMATED TEST PAGE 1 OF 1 DAMPER OUTPUT (MIN): 1 TEST TIME (MIN): 8 SELECT = START TEST	DAMPER TEST TEST ACTIVE COUNT DOWN TIMER (SEC): 60 DAMPER OUTPUT: ZONE 01 BACK = END TEST
DAMPER TEST TEST TYPE: AUTOMATED MANUAL SELECT = CONTINUE	MANUAL TEST PAGE 1 OF 2 DAMPER OUTPUT (MIN): 5 ZONE DAMPER: ZONE 01 SELECT = START TEST	MANUAL TEST PAGE 2 OF 2 DAMPER OUTPUT (MIN): 5 ZONE DAMPER: ZONE 01 SELECT = START TEST	DAMPER TEST TEST ACTIVE COUNT DOWN TIMER (SEC): 30 DAMPER OUTPUT: ZONE 01 BACK = END TEST
DAMPER TEST TEST ACTIVE COUNT DOWN TIMER (SEC): 300 DAMPER OUTPUT: ZONE 01 BACK = END TEST			

A1353-001

Figure 60: Damper test

Zone system faults

You can view zone system faults and clear fault logs on the Hx 3 Zone Control Module. View and clear faults for all zones using the Primary Zone Module (zones 1–8). View and clear faults for zones 5–8 using the Secondary Zone Module. See Figure 61 for an example of viewing and clearing faults on the Primary Zone Module.

To view and clear system faults:

- 1. On the **Zone Settings** screen, use the **Up** and **Down** buttons to navigate to the **Zone System Faults** option and press **Select**.
- 2. If using the Primary Zone Module, use the **Up** and **Down** buttons to navigate to the **Primary Zone** or **Secondary Zone** option, depending on which fault logs you want to view, and press **Select**.

Or

If using the Secondary Zone Module, proceed to step 3.

- 3. On the **Zone System Faults** screen, use the **Up** and **Down** buttons to navigate to the **View Stored Faults** option and press **Select**.
- 4. On the **Stored Faults** screen, view the fault log.
- 5. Press **Back** to return to the **Zone System Faults** screen.
- 6. Use the **Up** and **Down** buttons to navigate to the **Clear Stored Faults** option and press **Select**.
- 7. On the **Clear Stored Faults** screen, press **Select** to confirm you want to clear the fault log.
- 8. Press **Back** to return to previous screens as needed.

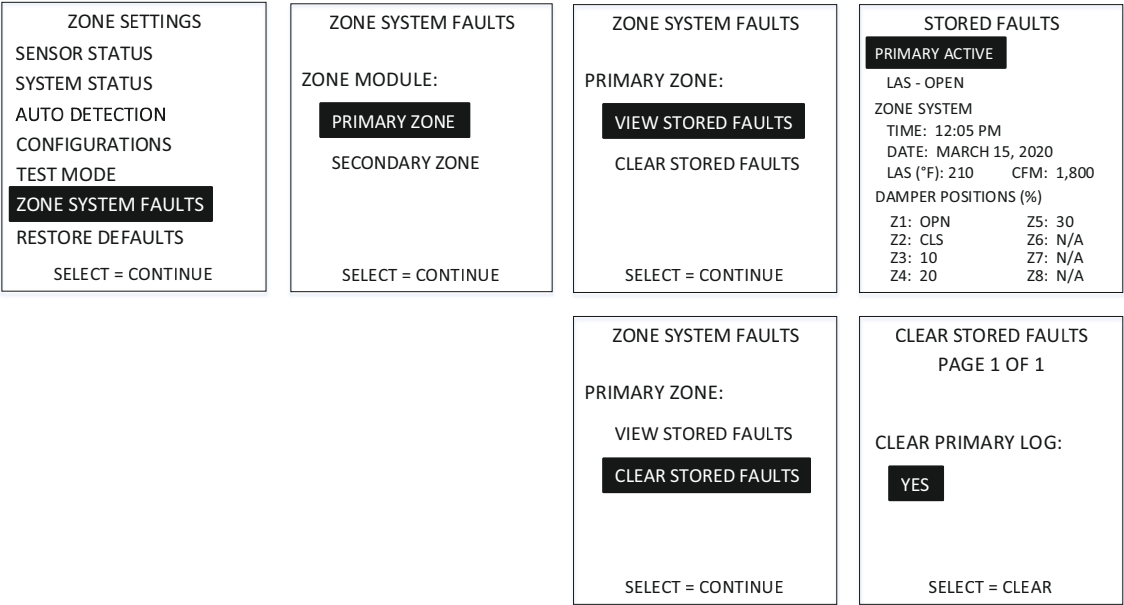


Figure 61: Primary Zone Module - zone system faults

Restore defaults



WARNING

The **Restore Defaults** option clears all settings configured by homeowners and installers.

Use the **Restore Defaults** option to clear all HVAC and zone system settings configured by installers and homeowners and revert to the default system settings. To do so:

1. On the **Zone Settings** screen, use the **Up** and **Down** buttons to navigate to the **Restore Defaults** option and press **Select**.
2. On the **Restore Sys Defaults** screen, press and hold **Select** for 5 s to confirm you want to perform a system restore.

If you do not want to perform a system restore, press **Back** to return to previous screens as needed.

Note: You can also restore the system defaults by pressing and holding the **Select** button on the Hx 3 Zone Control Module.



Figure 62: Primary Zone Module - restore system defaults



Figure 63: Secondary Zone Module - restore system defaults

Section VIII: Repair

If the system has been configured, restore the system defaults before replacing a zoning component or removing a zoning component from the system. Restoring the system defaults ensures that all system data across the HVAC Modbus and the zone Modbus is aligned.

Use the Main Control to access a list of installer settings. To do so, go to **System Settings > Restore System Defaults** and tap the **Installer Settings** icon. See Figure 64. This list contains current installer equipment settings, that is, settings configured after completion of the auto setup process. The list does not include settings configured by the homeowner, so note the homeowner settings separately. It is advisable to record the installer and homeowner settings before restoring the system defaults. Use the Restore defaults table in the Quick Reference Guide included in the Hx 3 Zone Control Module kit to record installer settings.

If the component you are replacing is a zone thermostat, configure the address of the new zone thermostat before performing the replacement.

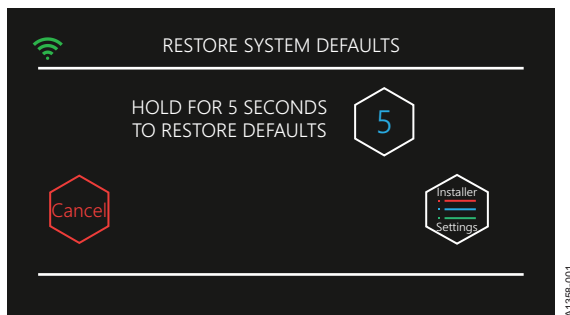


Figure 64: Hx 3 Touch Screen Thermostat (Main Control) - Restore system defaults

Restoring system defaults

Before replacing or removing a zoning component, use one of the following methods to restore the system defaults:

- Use the Hx 3 Touch Screen Thermostat acting as the Main Control: Go to **System Settings > Restore System Defaults**. On the **Restore System Defaults** screen, tap and hold for 5 s. See Figure 64.
- Use the **Restore Defaults** option on the Primary Zone Module or the Secondary Zone Module (if applicable): Follow the steps outlined in Restore defaults in Section VII: Hx 3 Zone Control Module - LCD and features.
- On the Primary Zone Module or the Secondary Zone Module (if applicable), press and hold the **Select** button.

Note: When the system defaults are restored, ensure the Main Control, zone thermostats (if applicable), the Primary Zone Module, and the Secondary Zone Module (if applicable) show preconfiguration screens before turning off system power and removing or changing the component. For example:

- Ensure the Main Control shows the **Auto Setup** screen. See Figure 11.
- Ensure zone thermostats show the **Zone Address** screen with addresses to be configured. See Figure 14.
- Ensure the Primary Zone Module and the Secondary Zone Module (if applicable) show the **Profile Incomplete** screen. See Figure 46.

Configuring zone thermostat addresses

Before removing and replacing an existing zone thermostat, configure the zone address of the new zone thermostat. This ensures the data of the new zone thermostat is passed across the HVAC Modbus and the zone Modbus. Configure the zone thermostat address using the **Zone Address** screen. See Figure 14.